

Why Rotating Fluids Make Polygons

Paper II: Classical and Quantum Physics

Gordon Speagle

Abstract

Building on the mathematical framework of Paper I, we derive the constrained energy landscape for point vortex rings, the central vortex stabilization mechanism, the $N = 7$ bifurcation, the blob bridge to continuous vorticity, the K_0 finite deformation radius interaction, circulation disorder resilience, and predictions for BEC vortex clusters on hyperbolic lattices.

1 Point vortex dynamics on a two-dimensional surface

On a two-dimensional surface, the stream function ψ satisfies the quasi-geostrophic potential vorticity (QGPV) equation

$$\frac{Dq}{Dt} = 0, \quad q = \nabla^2 \psi + \beta y. \quad (1.1)$$

The Green's function of the 2D Laplacian is $G(r) = -\ln r/(2\pi)$, satisfying $G'(r) = -1/(2\pi r) < 0$ for all $r > 0$.

For N point vortices with equal circulations κ on the collision-free configuration space $\{|z_j - z_k| > 0 \text{ for all } j \neq k\}$, where H is smooth and $\mathbb{Z}_N$ -invariant, the interaction energy is

$$H = - \sum_{j < k} \ln |z_j - z_k|, \quad (1.2)$$

where we absorb $\kappa^2/(4\pi)$ into the normalization. By the equivariant critical point theorem, the regular N -gon—a fixed point of $\mathbb{Z}_N$ —is a critical point of H .

Point vortex dynamics conserves three quantities: the energy H , the angular impulse $L = \sum |z_k|^2$, and the linear impulse $\mathbf{P} = \sum z_k$. The physically accessible configuration space is the $(2N - 3)$ -dimensional surface $\{L = NR^2, \mathbf{P} = 0\}$.

1.1 The centered spiral deformation

Consider N vortices on a ring of radius R , deformed along the one-parameter family

$$z_k(\sigma) = R e^{\sigma(k-(N-1)/2)/N} e^{2\pi i k/N}, \quad k = 0, \dots, N-1. \quad (1.3)$$

At $\sigma = 0$: the regular N -gon. The centering ensures $H(\sigma) = H(-\sigma)$ by the symmetry $k \mapsto N-1-k$ (which sends $\sigma \rightarrow -\sigma$ while preserving H), so $H'(0) = 0$. The spiral

deformation has a precise Möbius-group origin: the positions $\{z_k(\sigma)\}$ are the orbit of the seed $z_0 = R$ under the cyclic group generated by $z \mapsto \lambda z$, $\lambda = e^{\sigma/N + 2\pi i/N}$. At $\sigma = 0$ this generator is purely elliptic (rotation by $2\pi/N$, recovering the regular N -gon); for $\sigma \neq 0$ it is loxodromic ($|\lambda| \neq 1$) and the orbit traces the logarithmic spiral $r = R e^{(\sigma/2\pi)\theta}$ at angles $\theta = 2\pi k/N$. The spiral deformation is therefore the generator of the loxodromic one-parameter subgroup through the N -gon in the Möbius group's action on configuration space. $\mathbb{Z}_N$ -equivariant perturbations of the N -gon form a vector space; the spiral direction is singled out within this space as the unique direction tangent to a Möbius orbit (generated by a single transformation $z \mapsto \lambda z$ acting on a seed point and producing all N positions via the $\mathbb{Z}_N$ orbit). Other perturbation directions (individual Fourier modes, radial displacements) do not arise from a single Möbius seed and are analyzed through the full constrained Hessian in Theorem 2.2. The Möbius interpretation does not enter the stability proofs: the stability analysis uses the spiral only as a specific perturbation direction (the unique $\mathbb{Z}_N$ -equivariant direction that leaves the constraint surface at $O(\sigma^2)$, so that the unconstrained $H''(0) < 0$ does not contradict the constrained energy minimum). However, Proposition Paper I, Proposition 2.3 does carry genuine mathematical content beyond geometric decoration. Restricted to the dilation subgroup $f(z) = bz$, where f' is the constant b , the single-site correction becomes position-independent and reduces to $H(bz_1, \dots, bz_N) = H(z_1, \dots, z_N) - \frac{N(N-1)}{2} \ln b$. This is the N -body statement that H is dilation-invariant—the logarithmic energy changes only by an additive constant under spatial rescaling (Corollary Paper I, Corollary 2.4 above).

2 The constrained energy landscape

2.1 Energy curvature along the spiral

Theorem 2.1 (Energy curvature at the polygon). *For $N \geq 3$ vortices on a ring with $h(r) = -\ln r$, the energy along the centered spiral deformation satisfies $H'(0) = 0$ and*

$$H''(0) = -\frac{1}{4N^2} \sum_{m=1}^{N-1} \frac{(N-m)m^2}{\sin^2(\pi m/N)} < 0. \quad (2.1)$$

Proof. Part (a): $H'(0) = 0$ by the centering symmetry $k \mapsto N-1-k$. Part (b): Define $F(u, m) = \ln[1 - 2\cos(2\pi m/N) e^{um} + e^{2um}]$ with $u = \sigma/N$. Direct differentiation gives $F''(0, m) = m^2/[2\sin^2(\pi m/N)]$. There are $(N-m)$ pairs with separation m , and the chain rule $d/d\sigma = (1/N)d/du$ contributes $1/N^2$. This derivation uses the specific structure of $h = -\ln r$: the general formula Paper I, eq. 3.1 has two terms, but for the logarithmic interaction $(rh')' = (-1)' = 0$, so the S_m terms (first term in Paper I, eq. 3.1) vanish identically. Only the second term survives, giving (2.1) as a special case of Paper I, eq. 3.1 with $A_m = 0$ and $B_m = h'(d_m) m^2(N-m)/(N^2 d_m) = -m^2(N-m)/(N^2 d_m^2)$. Substituting $d_m = 2\sin(\pi m/N)$ yields (2.1). Every factor in the sum is strictly positive, so $H''(0) < 0$. $\square$

2.2 The constrained energy minimum

Attribution note. The Havelock eigenvalue formula $\lambda_m = (N-1) - m(N-m)/2$ and the linear stability boundary $N \leq 7$ are due to Havelock [1931]. The new contribution of the theorem below is the reframing: the N -gon is a *constrained energy minimum* (not a

saddle point or maximum), which resolves the apparent tension with Onsager's statistical-mechanics picture.

Theorem 2.2 (Constrained energy minimum). *The regular N -gon minimizes H on the conserved-quantity surface $\{L = NR^2, \mathbf{P} = 0\}$ if and only if $N \leq 7$. The Lagrangian Hessian*

$$\nabla^2 \mathcal{L} = \nabla^2 H - 2\mu_L I, \quad \mu_L = -\frac{N-1}{4}, \quad (2.2)$$

restricted to the constraint tangent space, is positive semi-definite for $N \leq 7$ and indefinite for $N \geq 8$.

Proof. The energy-Casimir functional is $\mathcal{H} = H + |\mu_L| L$, whose Hessian is (2.2). The Hessian $\nabla^2 H$ is a $2N \times 2N$ real symmetric matrix that commutes with the $\mathbb{Z}_N$ action $z_k \mapsto z_{k+1 \pmod{N}}$ (since H is invariant under cyclic permutation of equal-circulation vortices). By Schur's lemma for real representations of finite abelian groups, $\nabla^2 H$ block-diagonalizes in the discrete Fourier basis into 2×2 blocks, one per mode $m \in \{0, \dots, N-1\}$. For mode m , decompose the perturbation at vortex k into radial (δr_k) and tangential (δt_k) components, with Fourier amplitudes $\delta r_k = a \cos(2\pi m k/N)$, $\delta t_k = b \cos(2\pi m k/N)$. The 2×2 Hessian block for the pair (a, b) involves two sums over separations $p = 1, \dots, N-1$:

$$\begin{aligned} H_{rr}^{(m)} &= \sum_p \left[\frac{1 - \cos(2\pi p m/N)}{4 \sin^2(\pi p/N)} + \frac{1 + \cos(2\pi p m/N)}{4 \sin^2(\pi p/N)} \right] - 2 \sum_p \frac{1 - \cos(2\pi p m/N)}{4 \sin^2(\pi p/N)}, \\ H_{rt}^{(m)} &= \sum_p \frac{\sin(2\pi p m/N)}{4 \sin^2(\pi p/N)}, \\ H_{tt}^{(m)} &= -H_{rr}^{(m)}. \end{aligned}$$

The off-diagonal $H_{rt}^{(m)}$ vanishes identically by the anti-symmetry $\sin(2\pi p m/N) \rightarrow -\sin(2\pi(N-p)m/N)$ under $p \rightarrow N-p$, since $\sin^2(\pi p/N) = \sin^2(\pi(N-p)/N)$ (so the denominator is symmetric while the numerator is anti-symmetric; the sum over the full range $p = 1, \dots, N-1$ cancels in pairs). Therefore the block is diagonal: $\text{diag}(H_{rr}^{(m)}, -H_{rr}^{(m)})$, giving eigenvalues $\pm H_{rr}^{(m)} = \pm \sigma_m$.

The diagonal entry $H_{rr}^{(m)}$ simplifies using the Havelock identity (Appendix Paper I, §4.1): the self-sum $\sum_p 1/[4 \sin^2(\pi p/N)] = (N-1)/2$ (a special case of $T_0 + (N-1)/2$) and the cross-sum is $T_m = m(N-m)/2$. Thus

$$\sigma_m = H_{rr}^{(m)} = \frac{N-1}{2} - \frac{m(N-m)}{2}, \quad (2.3)$$

using

$$T_m := \sum_{p=1}^{N-1} \frac{1 - \cos(2\pi p m/N)}{4 \sin^2(\pi p/N)} = \frac{m(N-m)}{2} \quad (2.4)$$

(proven in Appendix Paper I, §4.1; cf. Havelock [1931], §3). The Lagrangian eigenvalues are thus

$$\lambda_m^+ = (N-1) - \frac{m(N-m)}{2}, \quad \lambda_m^- = \frac{m(N-m)}{2}.$$

Since $\lambda_m^- \geq 0$ always, positive semi-definiteness requires $\lambda_m^+ \geq 0$ for the modes tangent to the constraint surface. Three modes are removed: *Mode* $m = 0$ corresponds to uniform dilation $\delta z_k \propto z_k$, which changes $L = \sum |z_k|^2$ and is therefore not tangent to $\{L = NR^2\}$. *Modes* $m = 1$ and $m = N-1$ generate infinitesimal translation ($\delta z_k = \text{const}$, changing $\mathbf{P} = \sum z_k$) and overall rotation ($\delta z_k = i z_k$, a Goldstone mode of the $SO(2)$ symmetry of

H). Both have $\sigma_m = 0$, consistent with the continuous symmetry group; they lie in the kernel of $\nabla^2 H$ and produce the zero eigenvalues. The remaining modes $m = 2, \dots, N-2$ are tangent to the constraint surface and carry the stability information. The minimum of λ_m^+ over m occurs at $m = \lfloor N/2 \rfloor$:

$$\min_m \lambda_m^+ = \begin{cases} (N-1)(7-N)/8 & N \text{ odd,} \\ (N-1) - N^2/8 & N \text{ even.} \end{cases}$$

Both expressions are > 0 for $N \leq 6$, equal to zero at $N = 7$, and < 0 for $N \geq 8$. Thus $N \leq 6$ is *strictly* stable (positive definite Lagrangian), $N = 7$ is *neutrally* stable (positive semi-definite, with a zero eigenvalue at $m = 3$), consistent with Havelock [1931], and $N \geq 8$ is unstable. (The $N = 7$ marginal case is further analyzed in Section 4.) $\square$

Corollary 2.3 (Constrained minimum versus unconstrained saddle). *For $N \leq 7$, the N -gon is a constrained energy minimum on $\{L = NR^2, \mathbf{P} = 0\}$, yet a saddle point of the unconstrained energy H . These two facts are compatible: the constrained minimum is the relevant concept for Lyapunov stability via Arnold’s energy-Casimir method, while the Onsager negative-temperature argument operates on the unconstrained energy landscape, where the polygon is a maximum-energy accessible state (not a local maximum of H , but the highest- H configuration reachable within the Thomson-stable set). The distinction matters because the statistical mechanics literature sometimes describes the polygon as an energy maximum without qualification; it is a constrained minimum that happens to have higher energy than alternative configurations at the same conserved quantities.*

Proof. The unconstrained Hessian $\nabla^2 H$ has $H_{rr}^{(m)} > 0$ and $H_{tt}^{(m)} < 0$ at the N -gon: radial perturbations raise the energy while tangential perturbations lower it (the polygon is a saddle point of the unconstrained H , consistent with Onsager’s picture of local maxima in the energy landscape for same-sign vortices). The Lagrange multiplier $\mu_L = -(N-1)/4$ shifts all eigenvalues of $\nabla^2 H$ by $+2|\mu_L| = (N-1)/2$, converting the saddle into a minimum on the constraint surface. The value of μ_L follows from the constrained stationarity condition: at vortex 0 (position $(R, 0)$), the pair $(0, k)$ contributes identically $-1/(2R)$ to $\partial H/\partial x_0$ regardless of k , because $(1 - \cos \theta_k)/[4 \sin^2(\theta_k/2)] = 1/2$ for all k —a consequence of the scale invariance of $h'(r)/r = -1/r^2$. Summing over $N-1$ pairs gives $\partial H/\partial x_0 = -(N-1)/(2R)$, and setting this equal to $\mu_L \cdot 2R$ yields $\mu_L = -(N-1)/(4R^2)$.

In particular, Theorem 2.1 ($H''(0) < 0$ along the spiral direction) does not contradict Theorem 2.2: the spiral deformation (1.3) leaves the conserved surface at $O(\sigma^2)$, so it is not a constrained perturbation. The negative spiral curvature is an artifact of the angular impulse changing along the spiral path. $\square$

3 Central vortex stabilization

When a central vortex of circulation κ_0 is present at the origin, the central interaction $-\kappa_0 \sum_k \ln |z_k|$ modifies both the Lagrange multiplier and the Hessian. Two effects contribute to the critical eigenvalue:

Lagrange multiplier shift. The stationarity condition gives $\mu_L = -(N-1)/4 - \kappa_0/2$ (for $R = 1$), so the shift $-2\mu_L = (N-1)/2 + \kappa_0$ increases by $+\kappa_0$ per unit.

Hessian projection. The central-vortex Hessian at vortex k is the traceless 2×2 matrix $(\kappa_0/R^2) \begin{bmatrix} \cos 2\theta_k & \sin 2\theta_k \\ \sin 2\theta_k & -\cos 2\theta_k \end{bmatrix}$, which has eigenvalue $+\kappa_0/R^2$ for radial perturbations

$(\cos \theta_k, \sin \theta_k)$ and $-\kappa_0/R^2$ for tangential. For $N \geq 8$, the critical eigenvector of the ring-only Lagrangian (the most negative eigenvalue direction, at mode $m = \lfloor N/2 \rfloor$) is *purely radial* at every vortex position. This requires the off-diagonal block $H_{rt}^{(m)}$ to vanish at the critical mode $m = \lfloor N/2 \rfloor$, which we now verify explicitly. From equation (2.3): $H_{rt}^{(m)} = \sum_{p=1}^{N-1} \sin(2\pi pm/N)/[4\sin^2(\pi p/N)]$. The substitution $p \rightarrow N-p$ sends $\sin(2\pi pm/N) \rightarrow \sin(2\pi(N-p)m/N) = -\sin(2\pi pm/N)$ (anti-symmetric) while $\sin^2(\pi p/N) \rightarrow \sin^2(\pi(N-p)/N) = \sin^2(\pi p/N)$ (symmetric). Therefore each pair $(p, N-p)$ cancels, and $H_{rt}^{(m)} = 0$ for *every* m —in particular for the critical mode $m = \lfloor N/2 \rfloor$. The Hessian block is diagonal at all modes: $\text{diag}(H_{rr}^{(m)}, -H_{rr}^{(m)})$, so radial and tangential perturbations decouple completely. This decoupling is specific to the logarithmic (harmonic) interaction, for which $H_{tt} = -H_{rr}$. Therefore $\langle v_{\text{crit}} | \nabla^2 H_{\text{center}} | v_{\text{crit}} \rangle = +\kappa_0$.

The total shift of the critical eigenvalue is $+2\kappa_0$ per unit κ_0 (one from the Lagrange multiplier, one from the Hessian projection). Setting the shifted eigenvalue to zero:

Proposition 3.1 (Critical central vortex strength). *For $N \geq 8$, the N -gon with central vortex is a constrained energy minimum if and only if $\kappa_0/\kappa \geq \kappa_{\text{crit}}(N)$, where*

$$\kappa_{\text{crit}}(N) = \begin{cases} (N^2 - 8N + 8)/16 & N \text{ even}, \\ (N-1)(N-7)/16 & N \text{ odd}. \end{cases} \quad (3.1)$$

Proof. By Theorem 2.2, the minimum constrained Lagrangian eigenvalue at $\kappa_0 = 0$ is $\lambda_{\text{min}} = (N-1) - \lfloor N/2 \rfloor(N - \lfloor N/2 \rfloor)/2$. The central vortex shifts this eigenvalue by $+2\kappa_0$: $+\kappa_0$ from the Lagrange multiplier (since $-2\mu_L$ increases by κ_0) and $+\kappa_0$ from the central-vortex Hessian projection (since the critical eigenvector is purely radial, and the central-vortex Hessian has eigenvalue $+\kappa_0/R^2$ on radial perturbations). Setting $(N-1) - m(N-m)/2 + 2\kappa_0 = 0$ with $m = \lfloor N/2 \rfloor$ gives (3.1). $\square$

The first values are:

N	≤ 7	8	9	10	11	12	13	14
κ_{crit}	0	$\frac{1}{2}$	1	$\frac{7}{4}$	$\frac{5}{2}$	$\frac{7}{2}$	$\frac{9}{2}$	$\frac{23}{4}$

These agree with the classical Thomson dynamical stability analysis [Thomson, 1883, Havelock, 1931], confirming that energy and dynamical stability criteria are equivalent.

For $N = 6$, the exact critical ratio $\kappa_{\text{crit}}(6) = -1/4$ follows algebraically from the shifted-eigenvalue formula in the proof of Proposition 3.1. With $N = 6$ (even) and critical mode $m = \lfloor N/2 \rfloor = 3$, the Lagrangian eigenvalue with central vortex at strength $r = \kappa_0/\kappa$ is

$$\lambda_3 = (N-1) - \frac{m(N-m)}{2} + 2r = 5 - \frac{9}{2} + 2r = \frac{1}{2} + 2r.$$

Setting $\lambda_3 = 0$ gives $r = -1/4$. The closed form $\kappa_{\text{crit}} = (N^2 - 8N + 8)/16$ of Proposition 3.1 evaluates at $N = 6$ to $(36 - 48 + 8)/16 = -4/16 = -1/4$, consistent with this. (The Proposition is stated for $N \geq 8$; at $N = 6$ the formula returns $\kappa_{\text{crit}} < 0$, meaning destabilisation requires an anticyclonic central vortex with $r < 0$.)

The same threshold appears in the dynamical stability analysis. The 12×12 linearisation of the hexagonal ring with central vortex [Havelock, 1931] has a characteristic polynomial containing the low-frequency factor

$$9 + 36\kappa_0/\kappa + 16\pi^2\lambda^2.$$

The constant term satisfies $9 + 36r = 18\lambda_3$, so the dynamical condition $9 + 36r = 0$ is algebraically equivalent to $\lambda_3 = 0$, both giving $r = -1/4$. For $\kappa_0/\kappa > -1/4$ the hexagonal configuration is stable; for $\kappa_0/\kappa < -1/4$ an anticyclonic central vortex destabilises it.

The center-vs-ring geometry is essential: multi-ring configurations are generically less stable (Section 5).

4 The $N = 7$ bifurcation

At $N = 7$, the minimum constrained Lagrangian eigenvalue is zero (mode $m = 3$). In the language of critical phenomena (Remark Paper I, Remark 4.2), this is the critical point of a Landau mean-field transition: the spectral gap $\Delta(N) = C_1 - f_{\max}(N)$ closes linearly as $\Delta \sim -(N-7)$ with mean-field exponent $\nu = 1$, and the bifurcation is classified by the quartic term of H along the $m = 3$ neutral direction on the constraint surface. The $\mathbb{Z}_7$ -equivariant normal form is the Landau potential $V(\varepsilon) = (\Delta/2)\varepsilon^2 + (\alpha_0/4)\varepsilon^4$, where $\varepsilon = |z|$ is the amplitude of the $m = 3$ mode and $\Delta = 0$ at $N = 7$. Numerical evaluation (tracing H along the Newton-projected constraint surface) gives $\alpha > 0$. The neutral subspace of the constrained Hessian consists of two modes: the radial Fourier modes at $m = 3$ and $m = 4 = N-3$ (together forming the 2-dimensional space of symmetry-breaking perturbations), plus the rotation mode (trivially neutral by the $\text{SO}(2)$ symmetry of H).

For both neutral modes, the *unconstrained* quartic d^4H/dt^4 (where t is the unit-eigenvector amplitude) takes the same value by $\mathbb{Z}_7$ symmetry. Each of the 21 vortex pairs contributes

$$\frac{6C^2}{A^2} - \frac{12CB^2}{A^3} + \frac{3B^4}{A^4},$$

where $A = |\Delta z|^2$, $B = 2 \operatorname{Re}(\Delta z \cdot \overline{\Delta u})$, $C = |\Delta u|^2$ are evaluated at the unit-ring N -gon ($|\Delta z|^2 = 4 \sin^2(\pi p/N)$ for separation class p). Direct summation over all 21 pairs gives

$$\left. \frac{d^4H}{dt^4} \right|_{m=3} = \frac{153}{7}. \quad (4.1)$$

The quartic sum is a polynomial in coordinates of the N -gon (whose positions are 7th roots of unity) and is $\mathbb{Z}_7$ -invariant: simultaneously shifting all vortex labels by one maps the ring to itself and maps u to a rotated copy. By Galois theory, such an invariant sum lies in $\mathbb{Q}(\zeta_7)^{\mathbb{Z}_7} = \mathbb{Q}$, so the result is necessarily a rational number. The value $153/7$ is then identified by the following argument: the 21 pairs decompose into 6 separation classes $p = 1, \dots, 6$ with 7 pairs each, and the per-pair quartic (4.1) depends on p only through $A_p = 4 \sin^2(\pi p/7)$ and the mode overlap $B_p = 2 \operatorname{Re}(\Delta z \cdot \overline{\Delta u})$, $C_p = |\Delta u|^2$. These are algebraic numbers in $\mathbb{Q}(\zeta_7)$. The Galois rationality argument is the proof: $\mathbb{Z}_7$ -invariant sums over $\mathbb{Q}(\zeta_7)$ lie in the fixed field $\mathbb{Q}$, so the quartic coefficient is rational *a priori*; direct algebraic summation (`n7_bifurcation.py`, independently confirmed by 50-decimal-place evaluation) identifies the value as $153/7$. (The $\mathbb{Z}_7$ symmetry forces the degree-4 form on the 2-dimensional neutral subspace to be proportional to $|z|^4$, so $153/7$ holds for *any* unit vector in the $\{m=3, m=4\}$ subspace.)

The constraint-surface correction admits an exact closed form via the *log-scaling identity*. The Newton projection onto $\{L = N\}$ sends $\varepsilon \mapsto z(\varepsilon) = (z + \varepsilon u) \sqrt{N/L(\varepsilon)}$, where $L(\varepsilon) = N + \varepsilon^2$ exactly (since $\operatorname{Re} \sum_k z_k \overline{u_k} = 0$ for the radial $m=3$ mode and $\|u\|^2 = 1$).

Because the logarithmic energy is a sum of $-\ln|\cdot|$ over $N(N-1)/2$ pairs, the overall scale factor $s(\varepsilon) = \sqrt{N/(N+\varepsilon^2)}$ contributes an additive term:

$$H_{\text{constrained}}(\varepsilon) = H_{\text{unconstrained}}(\varepsilon) + \frac{N(N-1)}{4} \ln\left(1 + \frac{\varepsilon^2}{N}\right). \quad (4.2)$$

Expanding the logarithm: $\ln(1 + \varepsilon^2/N) = \varepsilon^2/N - \varepsilon^4/(2N^2) + O(\varepsilon^6)$, so the 4th derivative of the correction term at $\varepsilon = 0$ is

$$\frac{N(N-1)}{4} \cdot \left(-\frac{12}{N^2}\right) = -\frac{3(N-1)}{N}.$$

For $N = 7$ this is $-18/7$. Combined with the unconstrained value $153/7$, the constrained quartic is therefore

$$\left.\frac{d^4 H}{dt^4}\right|_{\text{constrained}} = \frac{153}{7} - \frac{18}{7} = \frac{135}{7}, \quad (4.3)$$

an exact rational result. Richardson extrapolation (three two-point pairs at $h \in \{0.05, 0.04, 0.03, 0.02\}$, symmetric fourth-difference formula $[2\delta H(2h) - 8\delta H(h)]/h^4 \rightarrow 24c_4$ with log-ratio cancellation) gives 19.2857 ± 0.0005 , numerically consistent with the exact value $135/7 = 19.285714\dots$

[Kurakin and Yudovich \[2002\]](#) established nonlinear stability of the heptagonal ring using Arnold's energy-Casimir method, proving positivity of the quartic form by Dirichlet's criterion without computing its value. [Cabral and Schmidt \[1999\]](#) and [Schmidt \[2004\]](#) independently proved stability via Birkhoff normal form and computed a quartic coefficient of $9/28$ in the *Sylvester-normalized* coordinates (where all quadratic terms are rescaled to $1/2$). The present value $\alpha_0 = 45/14$ is expressed in the physical (un-normalized) radial-mode coordinates on the constraint surface; both values confirm $\alpha_0 > 0$ but measure different quantities. The present derivation proceeds via the constrained energy-minimum framework (Theorem 2.2): the log-scaling identity (4.2) yields the exact correction $-3(N-1)/N$, and the unconstrained value $153/7$ is established by the Galois rationality argument combined with direct algebraic summation. In the normal form $\mathcal{H} = (\alpha_0/4)|z|^4$, the exact constrained quartic gives

$$\alpha_0 = \frac{135/7}{6} = \frac{45}{14} \approx 3.214 > 0: \quad (4.4)$$

the energy-Casimir functional $\mathcal{H} = H + |\mu_L|L$ has a strict local minimum along the $m = 3$ direction at quartic order, $\mathcal{H} \geq \mathcal{H}_0 + (\alpha_0/4)\varepsilon^4$. By the Dirichlet stability criterion [[Boatto and Cabral, 2003](#)], the heptagonal configuration is nonlinearly stable. The quartic (rather than quadratic) confinement explains why heptagonal rings are stable but more easily perturbed than hexagonal ones.

5 Multi-ring configurations and the center-ring dichotomy

Proposition 5.1 (Center-ring dichotomy). *Let N outer vortices sit at the regular N -gon on a ring of radius R_2 , and let $M \geq 2$ inner vortices sit at the regular M -gon on a ring of radius $R_1 < R_2$.*

(i) If $N \leq 7$ (outer ring Thomson-stable), the two-ring configuration is unstable: the Lagrangian Hessian has at least one negative eigenvalue.

(ii) If $N \geq 8$ (outer ring Thomson-unstable), the inner ring can reduce but not eliminate the instability: the two-ring Hessian retains at least one negative eigenvalue.

In contrast, a single central vortex ($M = 0$) at the $\mathbb{Z}_N$ fixed point produces a uniform eigenvalue shift $+\kappa_0/R^2$ across all Fourier modes and can achieve full stabilization (Proposition 3.1).

Proof. The proof splits into two cases.

Case 1: $N \geq 8$ (outer ring Thomson-unstable). The Havelock eigenvalue $\lambda_{\lfloor N/2 \rfloor} = (N-1) - \lfloor N/2 \rfloor(N - \lfloor N/2 \rfloor)/2 < 0$, so the outer N -ring alone has a negative constrained eigenvalue. Adding $M \geq 2$ inner vortices introduces new degrees of freedom but does not remove the existing instability: the negative outer mode persists (its eigenvalue is a continuous function of the coupling strength, starting negative at zero coupling). This proves (ii).

Case 2: $N \leq 7$ (outer ring Thomson-stable). The two concentric regular polygons form an equilibrium at the co-rotation ratio

$$R_2/R_1 = \sqrt{(N+M-1)/(M-1)}, \quad (5.1)$$

determined by the Gauss mean-value theorem: the outer ring at $R_2 > R_1$ encloses zero inner-ring circulation, so the inner ring's angular velocity is $(M-1)\kappa/(4\pi R_1^2)$ (self-interaction only), while the outer ring sees the enclosed inner circulation and rotates at $(N+M-1)\kappa/(4\pi R_2^2)$. Equating gives (5.1).

At this equilibrium, the constrained Lagrangian Hessian $\mathcal{H}$ is computed for all 12 cases ($N \in \{5, 6, 7\}$, $M \in \{2, \dots, N-1\}$). Non-positive-definiteness is certified by two algebraic tests:

- **Determinant test** (8 cases with odd number of negative eigenvalues): $\det(\mathcal{H}) < 0$ implies an odd number of negative eigenvalues, hence ≥ 1 .
- **Sylvester minor test** (4 remaining cases with even number of negative eigenvalues): the k -th leading principal minor of $\mathcal{H}$ is negative for some $k \leq \dim(\mathcal{H})$, so $\mathcal{H}$ is not positive semi-definite.

N	M	$\lambda_{\min}$	n_{neg}	Certificate
5	2	-0.375	3	$\det < 0$
5	3	-1.410	3	$\det < 0$
5	4	-3.085	4	$\text{minor}(7) < 0$
6	2	-0.204	2	$\text{minor}(8) < 0$
6	3	-0.795	3	$\det < 0$
6	4	-1.688	4	$\text{minor}(9) < 0$
6	5	-3.136	5	$\det < 0$
7	2	-0.088	1	$\det < 0$
7	3	-0.316	3	$\det < 0$
7	4	-0.803	3	$\det < 0$
7	5	-1.691	5	$\det < 0$
7	6	-3.012	6	$\text{minor}(9) < 0$

Both tests are verified in *exact rational arithmetic* (sympy 1.14, all matrix entries converted to $\mathbb{Q}$ via `nsimplify`; determinants computed symbolically with no floating-point rounding). The signs are algebraically exact: there is no numerical error to bound. Code: `two_ring.py` (numerical), `symbolic_two_ring_proof.py` (exact).

Extension to all $M \geq 2$. The 12 exact cases cover $M \in \{2, \dots, N-1\}$. The result extends to all $M \geq N$ by the following argument. The 2×2 radial Lagrangian Hessian (the breathing modes of the two rings) has off-diagonal entry $H_{12} = \partial^2 H / (\partial R_1 \partial R_2)$ from the attractive inter-ring interaction. At the co-rotation equilibrium, each ring individually has positive radial stiffness, but the shared Lagrange multiplier μ_L satisfies $\mu_{\text{outer}} < \mu_L < \mu_{\text{inner}}$ (the inner ring's natural frequency exceeds the outer's by the ratio $(N-1)/(N+M-1) < 1$). The resulting determinant of the 2×2 radial block satisfies $\det = (H_{11} - 2M\mu)(H_{22} - 2N\mu) - H_{12}^2$. Since H_{12} is proportional to MN (each inner-outer pair contributes) and the diagonal terms scale as M^2 and N^2 , the off-diagonal dominates for $M \geq 2$: $\det < 0$, giving at least one negative eigenvalue. This is confirmed by the 12 exact cases ($\det < 0$ in all 12) and extends to $M \geq N$ by continuity in M (the determinant is a rational function of M with no sign change for $M \geq 2$).

Mechanism. A central vortex ($M = 0$, the $\mathbb{Z}_N$ fixed point) contributes a uniform eigenvalue shift $+\kappa_0/R^2$ across all Fourier modes, compensating the negative Havelock eigenvalues. An inner ring of $M \geq 2$ cannot achieve this: the inter-ring coupling H_{12} breaks the uniformity, and the Lagrange-multiplier frustration ($\mu_{\text{outer}} \neq \mu_{\text{inner}}$) forces a negative eigenvalue. $\square$

Jupiter's observed center-plus-octagon geometry is therefore *uniquely stabilized* among all configurations with a secondary vortex structure: only the single central cyclone achieves full stabilization of the $N = 8$ ring.

6 From point vortices to continuous vorticity: the blob bridge

The results of Sections 1.1–Paper III, §21 are stated for point vortices. The RG classification (Proposition Paper I, Proposition 2.1) guarantees that compact vorticity distributions interact logarithmically at large separation, with irrelevant corrections $O(\varepsilon^2/d^2)$. Two distinct bridges make this quantitative.

6.0.1 The vortex blob bridge (Jupiter regime)

For discrete cyclone configurations ($\beta R^3/\kappa \ll 1$), replace each point vortex with a smooth blob of width ε . The regularized interaction $h_\varepsilon(d) = -\frac{1}{2} \ln(d^2 + 2\varepsilon^2)$ recovers $-\ln d$ as $\varepsilon \rightarrow 0$.

Proposition 6.1 (Blob stability correction). *The constrained Lagrangian eigenvalues for h_ε satisfy*

$$\lambda_m(\varepsilon) = \lambda_m(0) + \left[-\frac{N^2-1}{12} + P_m\right] \varepsilon^2 + O(\varepsilon^4), \quad (6.1)$$

where $\lambda_m(0)$ is the Havelock eigenvalue (Theorem 2.2).

Proof. Lagrange multiplier shift. The stationarity condition $\partial H_\varepsilon / \partial x_0 = \mu_L \cdot 2x_0$ gives $\mu_L(\varepsilon) = \mu_L(0) + \delta\mu_L \cdot \varepsilon^2$ where $\delta\mu_L = (N^2-1)/24$, using $\sum_{k=1}^{N-1} \csc^2(\pi k/N) = (N^2-1)/3$. The shift $-2\delta\mu_L = -(N^2-1)/12$ is *universal* (the same for all Fourier modes).

Hessian correction. The $O(\varepsilon^2)$ corrections to h'' and h'/d are: $\delta(h''(d)) = -6\varepsilon^2/d^4 + O(\varepsilon^4)$, $\delta(h'(d)/d) = 2\varepsilon^2/d^4 + O(\varepsilon^4)$. For a pair at distance $d_p = 2\sin(\pi p/N)$, the Hessian correction per pair is $O(\varepsilon^2/d_p^4)$. The projection $P_m = v_m^\top \delta \nabla^2 H v_m$ of this correction onto the critical eigenvector v_m is computed by summing over all $N - 1$ pairwise distances with the Fourier mode weights.

Operator norm bound. Since $d_{\min} = 2\sin(\pi/N) > 0$:

$$\|\nabla^2 H_\varepsilon - \nabla^2 H_0\|_{\text{op}} \leq 6(2N-3)\varepsilon^2/d_{\min}^4 = O(\varepsilon^2). \quad (6.2)$$

By Weyl's perturbation theorem, each eigenvalue shifts by at most this amount, giving uniform convergence at $O(\varepsilon^2)$.

Profile independence. For a blob with finite second moment σ^2 (Gaussian, compact support), the Fourier transform $|\hat{\phi}(\varepsilon k)|^2 = 1 - \sigma^2 \varepsilon^2 |k|^2 + O(\varepsilon^4 |k|^4)$ shows the $O(\varepsilon^2)$ correction is a constant in k -space (= self-energy, not affecting pairwise eigenvalues). The pairwise correction is $O(\varepsilon^4)$ —*faster* than the Cauchy blob. Since the Cauchy blob is the slowest-converging regularization, all profiles with finite σ^2 inherit its stability conclusions. $\square$

The Hessian projection P_m and the total correction $c_m = -(N^2-1)/12 + P_m$ (specific to the Cauchy blob $h_\varepsilon = -\frac{1}{2}\ln(d^2+2\varepsilon^2)$) are exact rational numbers, computed as follows. Differentiating the Cauchy blob pair Hessian with respect to ε^2 at $\varepsilon = 0$ gives the per-pair correction:

$$\delta H_{jj,xx} = \frac{2}{d_{jk}^6}((\Delta y)^2 - 3(\Delta x)^2), \quad \delta H_{jk,xx} = -\delta H_{jj,xx}, \quad (6.3)$$

with analogous expressions for yy and xy (see proof of Proposition 6.1). Here $(\Delta x, \Delta y, d_{jk})$ are the pair separation components. The critical radial Fourier mode is $v_k \propto z_k \cos(2\pi m k/N)$ with $m = \lfloor N/2 \rfloor$. The projection

$$P_m = v_m^\top \delta \nabla^2 H v_m \quad (6.4)$$

is a finite quadratic form: a sum over $N(N-1)$ pair contributions, each involving trigonometric values at N -th roots of unity. Since $\sin(\pi p/N)$ and $\cos(2\pi p/N)$ are algebraic for integer p and N , every term is rational and so is the sum. Explicit values (verified to machine precision in `blob_correction.compute_Pm(N)` and notebook `05.BlobCorrection.ipynb`):

N	3	4	5	6	7	8
$-(N^2-1)/12$	$-\frac{2}{3}$	$-\frac{5}{4}$	-2	$-\frac{35}{12}$	-4	$-\frac{21}{4}$
P_m	$-\frac{1}{3}$	$+\frac{1}{4}$	$+1$	$+4$	$+8$	$+\frac{65}{4}$
c_m	-1	-1	-1	$+\frac{13}{12}$	$+4$	$+11$

The Hessian projection P_m changes sign between $N = 3$ ($P_3 = -1/3$) and $N = 4$ ($P_4 = +1/4$): for the triangle, the blob softening reduces the nearest-neighbor curvature (all pairs are at the same short distance $d = \sqrt{3}$), while for the square and larger polygons, the long-range pairs increasingly dominate the projection and $P_m > 0$. Despite this sign change, $c_m = -1$ universally for $N \leq 5$ because the Lagrange multiplier shift $-(N^2-1)/12$ dominates P_m in this range. For $N \geq 6$, P_m grows faster and overwhelms the shift.

For blob profiles with finite second moment σ^2 (Gaussian, compact support), the $O(\varepsilon^2)$ correction is confined to the self-energy and does not affect pairwise eigenvalues (Section 6, profile independence argument). The pairwise correction is $O(\varepsilon^4)$ with a

profile-dependent coefficient. Since the $O(\varepsilon^2)$ corrections from the Cauchy blob table above preserve stability for $N \leq 7$, and all finite- σ^2 blobs have *smaller* corrections at any given ε , the stability conclusions hold *a fortiori* for all smooth profiles.

Three consequences follow:

(i) *Stability preserved.* For $N \leq 5$: $\lambda_0 \geq 1$ and $c = -1$, so $\lambda(\varepsilon) = \lambda_0 - \varepsilon^2 > 0$ for $\varepsilon < 1$. For $N = 6$: $c > 0$, so $\lambda(\varepsilon) > \lambda_0$ —the blob *enhances* stability.

(ii) *Marginal mode resolved.* At $N = 7$, $\lambda_0 = 0$ and $c = +4$, so $\lambda(\varepsilon) = 4\varepsilon^2 > 0$ for any $\varepsilon > 0$. The heptagonal marginal stability is an artifact of the singular point vortex limit.

(iii) *Instability partially compensated.* For $N = 8$: $\lambda_0 = -1$ and $c = 11$, so the octagon is stabilized for $\varepsilon > 1/\sqrt{11} \approx 0.30$.

6.0.2 Full infinite-dimensional convergence

Proposition 6.1 addresses center-position perturbations of N smooth blobs. The full vorticity perturbation space is infinite-dimensional, including blob *shape* deformations. A complete stability analysis requires the Hessian of the 2D Euler energy $E[\omega] = \frac{1}{2} \iint \omega(x) G(x, y) \omega(y) dx dy$ at the blob configuration $\omega_\varepsilon = \sum_k \kappa \phi_\varepsilon(\cdot - z_k)$, decomposed into three sectors:

(a) *Center translations.* The Hessian restricted to the $2N$ -dimensional subspace of center displacements is *exactly* the blob Hessian of Proposition 6.1. Convergence to the point vortex Hessian at $O(\varepsilon^\alpha)$ follows from the convolution estimates.

(b) *Shape perturbations.* Mean-zero, center-preserving deformations of each blob's profile enter through the self-energy $E_{\text{self}} = \iint \phi_\varepsilon(x) G(x, y) \phi_\varepsilon(y) dx dy$. The second variation at mean-zero $\delta\omega$ is, by Plancherel and the Fourier representation $\hat{G}(\xi) = 1/(4\pi^2|\xi|^2)$ of the 2D log-Green's function:

$$\delta^2 E_{\text{self}} = \iint \delta\omega(x) G(x, y) \delta\omega(y) dx dy = \int_{\mathbb{R}^2} \frac{|\hat{\delta\omega}(\xi)|^2}{4\pi^2|\xi|^2} d\xi = \frac{1}{4\pi^2} \|\delta\omega\|_{H^{-1}}^2. \quad (6.5)$$

Since $\hat{\delta\omega}(0) = \int \delta\omega = 0$ (mean-zero), the integrand is zero only on a set of measure zero, so $\delta^2 E_{\text{self}} > 0$ for all non-zero mean-zero $\delta\omega$: the self-energy Hessian is positive definite in H^{-1} . The rescaling $\phi_\varepsilon(x) = \varepsilon^{-2} \phi(x/\varepsilon)$ gives $\hat{\phi}_\varepsilon(\xi) = \hat{\phi}(\varepsilon\xi)$, so

$$E_{\text{self}}[\phi_\varepsilon] = \int \frac{|\hat{\phi}(\varepsilon\xi)|^2}{4\pi^2|\xi|^2} d\xi = \varepsilon^{-2} \int \frac{|\hat{\phi}(\eta)|^2}{4\pi^2|\eta|^2} d\eta = \frac{c}{\varepsilon^2},$$

and the smallest shape eigenvalue scales as c/ε^2 . Shape modes are *unconditionally stable*.

(c) *Center-shape coupling.* The cross terms involve overlap integrals

$$C_{jk} = \iint \nabla \phi_\varepsilon(x - z_j) \cdot \nabla G(x, y) \delta\psi(y - z_k) dx dy$$

between non-adjacent blobs $j \neq k$ ($|z_j - z_k| \geq d_{\min} \gg \varepsilon$), where $\delta\psi$ is a mean-zero, center-preserving shape perturbation. Since both the monopole ($\int \delta\psi = 0$) and dipole ($\int y \delta\psi = 0$) moments of $\delta\psi$ vanish, the leading term in the multipole expansion of $G(x, y)$ around z_k is the quadrupole:

$$G(x, y) = G(x, z_k) + \nabla_y G(x, z_k) \cdot (y - z_k) + O(|y - z_k|^2/d^2), \quad d = |x - z_k|.$$

The first two terms contribute zero (by the vanishing moments), so the coupling is determined by the quadrupole interaction: $C_{jk} = O(\varepsilon^2/d_{\min}^3)$. (Numerically verified:

$C_{jk} \approx 4\varepsilon^2/d_{\min}^3$ for the Cauchy blob at $d_{\min} \geq 3\varepsilon$.) This is *algebraically small*—the correct bound for the Cauchy blob, which has algebraic not exponential spatial decay. The ratio of coupling to shape eigenvalue is $O(\varepsilon^4/d_{\min}^3) \rightarrow 0$ as $\varepsilon \rightarrow 0$ with $d_{\min}$ fixed, which is the only bound needed below.

Proposition 6.2 (Inertia preservation). *For sufficiently small $\varepsilon > 0$, the full Hessian of $E[\omega_\varepsilon] + \mu_L L[\omega_\varepsilon]$ on the space of $\mathbb{Z}_N$ -symmetric vorticity perturbations has the same inertia (number of positive, zero, and negative eigenvalues along the constrained directions) as the point vortex Lagrangian Hessian of Theorem 2.2. The point vortex model is therefore the exact limit of smooth vorticity dynamics for stability analysis: no eigenvalue is lost, no sign is changed, and no new instability arises from the infinite-dimensional shape degrees of freedom. The stability boundary $N \leq 7$ is not an approximation of the smooth result—it is the smooth result in the $\varepsilon \rightarrow 0$ limit.*

Proof. Write the constrained full Hessian in block form:

$$\mathcal{H} = \begin{pmatrix} H_{cc} & H_{cs} \\ H_{cs}^\top & H_{ss} \end{pmatrix},$$

where the subscripts c and s denote the center and shape sectors.

Shape sector. By (6.5), $H_{ss} \geq (c/\varepsilon^2) \text{Id}_{H^{-1}}$ with $c > 0$. All shape eigenvalues are bounded below by $c/\varepsilon^2 > 0$.

Center sector. By (a) and the convolution estimates of Proposition 6.1, the center-sector eigenvalues satisfy $\lambda_c^{(k)}(\varepsilon) = \lambda_k^{\text{Hav}} + O(\varepsilon^2)$, where λ_k^{Hav} are the Havelock eigenvalues.

Coupling bound. From (c), the off-diagonal block satisfies $\|H_{cs}\|_{\text{op}} = O(\varepsilon^2/d_{\min}^3)$. By the Schur complement, the full Hessian has the same inertia as $H_{cc} - H_{cs}H_{ss}^{-1}H_{cs}^\top$. The correction term is

$$\|H_{cs}H_{ss}^{-1}H_{cs}^\top\|_{\text{op}} \leq \|H_{cs}\|_{\text{op}}^2 \|H_{ss}^{-1}\|_{\text{op}} = O\left(\frac{\varepsilon^4}{d_{\min}^6}\right) \cdot O(\varepsilon^2) = O\left(\frac{\varepsilon^6}{d_{\min}^6}\right),$$

which goes to zero as $\varepsilon \rightarrow 0$ with $d_{\min}$ fixed. Hence, for sufficiently small ε , the Schur complement eigenvalues are within $O(\varepsilon^6/d_{\min}^6)$ of the center-sector eigenvalues H_{cc} , and in particular preserve their signs. Since $\|H_{cc} - H_{cc}^{\text{Hav}}\|_{\text{op}} = O(\varepsilon^2)$ (from the blob correction), for small enough ε the Schur-complement eigenvalues have the same signs as the Havelock eigenvalues. Therefore the full Hessian inertia equals the Havelock inertia—which is the point vortex Lagrangian inertia of Theorem 2.2. $\square$

The result extends to general $\mathbb{Z}_N$ -symmetric vorticity concentrated in N patches: the center-shape coupling is quadrupole ($O(\varepsilon^2/d_{\min}^2)$ for compact profiles), and the Schur correction is $O(\varepsilon^6/d_{\min}^6)$. The explicit concentration condition for sign preservation is

$$\frac{\varepsilon}{d_{\min}} < \delta(N) := \sqrt{\frac{\lambda_{\min}(N)}{6(2N-3)}}, \quad (6.6)$$

giving $\delta(4) = 0.18$, $\delta(5) = 0.15$, $\delta(6) = 0.096$ (set by the center-sector bound; the Cauchy blob, with the slowest algebraic decay, provides the worst case).

Corollary 6.3 (Inertia preservation for $N \leq 6$). *For $N \leq 6$ and any $\mathbb{Z}_N$ -symmetric vorticity distribution with patch radius $\varepsilon/d_{\min} < \delta(N)$ (equation (6.6)), the full Hessian inertia equals the Havelock inertia. The argument uses only the spectral-gap condition $\lambda_{\min}(N) > 0$ and the Weyl perturbation bound; no information about the $O(\varepsilon^2)$ blob correction is needed.*

Corollary 6.4 (Marginal-mode resolution for $N = 7$). *For $N = 7$, the concentration bound (6.6) does not apply ($\lambda_{\min} = 0$, so $\delta(7) = 0$). A separate, logically distinct argument shows $\lambda_{\min}(\varepsilon) > 0$ for all $\varepsilon \in (0, d_{\min})$ via two regions:*

- Region 1 ($\varepsilon < \varepsilon_0 = 2/\sqrt{|c'_7|}$, where $|c'_7| \leq 17$ is the bounded $O(\varepsilon^4)$ coefficient): $\lambda \geq \varepsilon^2(4 - |c'_7|\varepsilon^2) > 0$, since the blob correction $c_7 = +4 > 0$ (Proposition 6.1) dominates.
- Region 2 ($\varepsilon \in [\varepsilon_0, d_{\min})$): interval arithmetic on a 2,000-point grid with mean-value-theorem bounds gives $\lambda \geq 0.34 > 0$ on every subinterval.

The two corollaries rest on different mechanisms: Corollary 6.3 on a spectral gap, Corollary 6.4 on a blob correction. Both extend beyond prescribed-profile blobs to general $\mathbb{Z}_N$ -symmetric vorticity patches satisfying their respective conditions.

6.0.3 The Rossby bridge (Saturn regime)

For continuous jet meanders ($\beta R^3/\kappa \gg 1$), the bridge is not vortex concentration but *Rossby wave quantization*. The QGPV linearized about a zonal jet $U(y)$ gives the Rayleigh–Kuo equation:

$$(U - c)(\hat{\psi}'' - k^2\hat{\psi}) + (\beta - U'')\hat{\psi} = 0.$$

For a stationary wave ($c = 0$) at the jet center ($y = 0$) of a Gaussian jet with $U''(0) = -U_{\max}/\sigma^2$:

$$\hat{\psi}'' + \left[\frac{\beta_{\text{eff}}}{U_{\max}} - \frac{n^2}{R^2} \right] \hat{\psi} = 0, \quad \beta_{\text{eff}} = \beta + \frac{U_{\max}}{\sigma^2},$$

where the effective PV gradient β_{eff} includes the jet curvature (which dominates β by a factor ~ 13 for Saturn). The stationary azimuthal wavenumber is

$$n^* = R_S \cos \varphi \sqrt{\beta_{\text{eff}}/U_{\max}}, \quad (6.7)$$

where $R_S \cos \varphi$ is the ring radius at latitude φ ($\approx 1.31 \times 10^7$ m for Saturn at 76° N). The nearest integer $n = \text{round}(n^*)$ is the selected polygon wavenumber. For Saturn: $\beta_{\text{eff}}/U_{\max} \approx 1.72 \times 10^{-13} \text{ m}^{-2}$, giving $n^* \approx 5.4$ from the leading-order dispersion relation. The observed $n = 6$ is obtained in numerical eigenvalue computations of the full Rayleigh–Kuo equation [Sánchez-Lavega et al., 2014]; the discrepancy between the leading-order $n^* \approx 5.4$ and the observed $n = 6$ reflects corrections from the finite jet width (the meridional eigenfunction structure contributes an $O(1/\sigma^2)$ correction to the effective wavenumber that is not captured by the dispersion relation alone).

The discretization mechanism is *azimuthal periodicity*: the wave must fit an integer number of wavelengths around the ring.

The connection to the point vortex energy landscape proceeds through the *critical layer* of the saturated wave. At finite amplitude, the stationary Rossby wave develops a cat’s-eye structure at the critical line $U = c = 0$: vorticity is trapped into N lobes of width $\sim \sigma$ centered at the meander peaks. As the jet sharpens ($\sigma/R \rightarrow 0$ at fixed wave amplitude), these lobes concentrate into N isolated patches near the N -gon vertices.

Remark 6.1 (Cat’s-eye Hessian convergence). *The cat’s-eye vorticity decomposes as $\omega = \omega_{\text{lobe}} + \omega_{\text{braid}} + \omega_{\text{bg}}$, where the lobes are the trapped fluid inside the separatrices, the braids are thin filaments along the separatrix itself, and the background is the passing fluid outside the cat’s-eye.*

Two conditions are required for Proposition 6.2 to apply:

(i) **Lobe concentration.** The lobes satisfy the concentration condition (6.6) with $\varepsilon \sim \sigma$ and the centroids converge to the N -gon vertices. This is guaranteed by $\mathbb{Z}_N$ symmetry.

(ii) **Braid smallness.** The absolute braid circulation Γ_{braid} is negligible compared to the total lobe circulation Γ_{lobe} .

Both conditions hold for the Stuart vortex $\Psi = -\sigma^2 \ln(\cosh(y/\sigma) - \eta \cos(n\theta))$ ($\eta \in (0, 1)$; $\eta \rightarrow 1$ is point-vortex concentration).

Proof of lobe dominance. The Stuart vortex satisfies $\omega = C/F^2$ with $F = \cosh(y/\sigma) - \eta \cos(n\theta)$. Near the elliptic center ($\theta = 0$, $y = 0$): $F \approx (1-\eta) + (y/\sigma)^2/2 + n^2\eta\theta^2/2$, so

$$\Gamma_{\text{lobe}} = C \iint_{\text{lobe}} \frac{dA}{F^2} \geq C \int_0^\infty \frac{2\pi r dr}{[(1-\eta) + r^2/2]^2} = \frac{2\pi C}{1-\eta}.$$

Outside the separatrix, $F \geq 1+\eta \geq 1$, so $\Gamma_{\text{non-lobe}} = C \iint_{\text{outside}} F^{-2} dA \leq C A_{\text{period}}$, which is bounded independently of η . Therefore

$$\frac{\Gamma_{\text{non-lobe}}}{\Gamma_{\text{total}}} \leq \frac{A_{\text{period}}(1-\eta)}{2\pi + A_{\text{period}}(1-\eta)} = O(1-\eta). \quad (6.8)$$

The lobe fraction $\Gamma_{\text{lobe}}/\Gamma_{\text{total}} = 1 - O(1-\eta)$ as $\eta \rightarrow 1$. Numerically: at $\eta = 0.99$, lobes capture $> 99\%$ of the total circulation.

Combining with Proposition 6.2: the cat's-eye satisfies the blob bridge conditions for $\sigma/d_{\min} < \delta(N)/\sqrt{2}$ and $1 - \eta < c\delta(N)^2$ (the braid perturbation is $O(1-\eta)$ of the lobe system, negligible at the same threshold). The constrained Hessian inertia of the cat's-eye equals the Havelock inertia, verified numerically for all $\eta \in [0.3, 0.99]$ and confirmed analytically by (6.8) in the limit $\eta \rightarrow 1$. The same argument applies to the Prandtl-Batchelor cat's-eye (homogenised PV inside the lobes), the physically relevant model for Saturn's hexagon. Code: `catseye_decomposition.py`, `catseye_hessian.py`.

The Saturn bridge thus arrives at the same energy landscape as the Jupiter bridge, through a different limiting procedure: Saturn concentrates a continuous jet meander into N patches (jet width $\sigma \rightarrow 0$), while Jupiter concentrates N discrete cyclones into

7 Finite Rossby deformation radius: the K_0 interaction

The K_0 interaction provides a second example of the RG flow toward the logarithmic fixed point (Proposition Paper I, Proposition 2.1): the exponential tail $K_0(r/R_d) + \ln r = O(e^{-r/R_d})$ is an irrelevant perturbation (faster than any power law), and the limit $R_d \rightarrow \infty$ is the flow to the fixed point.

In quasi-geostrophic dynamics with finite Rossby deformation radius R_d , the Green's function is $K_0(r/R_d)$ (modified Bessel function), which recovers $-\ln r$ for $r \ll R_d$. The interaction $h(r) = K_0(r/R_d)$ satisfies $h'(r) = -K_1(r/R_d)/R_d < 0$ and $(rh'(r))' = rK_0(r/R_d)/R_d^2 > 0$ (class (iii)). Numerical computation of the constrained Lagrangian Hessian as a function of $\eta = R/R_d$ (polygon ring radius over deformation radius) reveals an N -dependent *polygon-suppression transition*:

N	Barotropic $\lambda_{\min}$	Critical R/R_d	Status
≤ 5	≥ 1	none	Stable for all R/R_d
6	$1/2$	≈ 1.29	Suppressed for $R/R_d > 1.29$
7	0	≈ 0	Unstable for any finite R/R_d
≥ 8	< 0	N/A	Unstable (barotropic)

Polygons with $N \leq 5$ are robust against stratification effects. The hexagon ($N = 6$) is the first polygon sensitive to R_d : it requires $R/R_d < 1.29$. The heptagon ($N = 7$), neutrally stable in the barotropic limit, is destabilized by any finite R/R_d —providing an additional mechanism for its rarity beyond the quartic confinement.

For Saturn, the barotropic deformation radius (~ 2000 – 3000 km) gives $R/R_d \approx 8$ – 12 at the hexagonal jet, placing it above the critical ratio. The observed hexagonal persistence therefore requires the dynamics to be effectively barotropic over a deep layer—a requirement now directly supported by observational data. Cassini Grand Finale gravity measurements constrain Saturn’s differential rotation to a depth of $8,000$ – $9,000$ km [Iess et al., 2019], implying a coherent barotropic column of order 10^4 km, far exceeding the hexagon’s vertical scale. Saturn is therefore the best-supported case in our sample for the applicability of the 2D logarithmic Green’s function: the columnar depth is observationally established, not merely assumed.

Observable prediction: hexagon suppression signatures. The critical ratio $R/R_d \approx 1.29$ for $N = 6$ yields a testable prediction. If the effective barotropic depth were reduced (by seasonal radiative forcing of the upper troposphere, or by a hypothetical change in deep convective coupling), R_d would decrease and R/R_d would increase toward the critical value. Approaching the threshold, the framework predicts three observable signatures: (1) the minimum Havelock eigenvalue $\lambda_{\min}(\eta) \rightarrow 0$, reducing the hexagonal mode’s restoring frequency and producing a *longer oscillation period* for meridional meander perturbations; (2) the hexagonal amplitude ε_{obs} would *grow*, since the restoring force weakens while convective forcing remains constant; and (3) at $R/R_d > 1.29$, the hexagonal mode becomes linearly unstable and the pattern would transition to a lower- N polygon ($N = 5$, the next Thomson-stable configuration) or break up entirely. Saturn’s hexagon has been observed continuously since Voyager 1 (1980–81); Cassini monitored it through northern spring equinox (2009) to summer solstice (2017) with no sign of weakening [Antuñano et al., 2015, Sánchez-Lavega et al., 2014]. This four-decade persistence across half a Saturnian year places an empirical lower bound on the effective barotropic depth: seasonal radiative forcing does not reduce R_d enough to approach the critical ratio, consistent with the deep ($\sim 10^4$ km) barotropic column implied by Cassini gravity.

Why the leading K_0 correction does not shift stability. The small- r expansion $K_0(r/R_d) = -\ln(r/R_d) - \gamma_E + r^2/(4R_d^2) + O(r^4)$ shows the leading correction to the logarithm is $g(r) = r^2/(4R_d^2)$ —a perturbation of RG exponent $\alpha = 2$. By the isotropic-marginality result (Paper I, Proposition 2.2), the pair Hessian anisotropy $g'' - g'/r = \alpha(\alpha-2)r^{\alpha-2}$ vanishes at $\alpha = 2$: the Hessian perturbation is purely isotropic ($\propto I$) and is exactly compensated by the rotation-frequency shift $\delta\Omega$. Consequently $\delta\lambda_m = 0$ for all binding modes—the $r^2/(4R_d^2)$ term shifts the precession rate of the polygon but does *not* shift its stability eigenvalues. The first correction to stability from K_0 is $O(r^4/R_d^4)$ ($\alpha = 4$, which is *not* isotropic-marginal), and this is numerically small for $R/R_d \ll 1$.

This explains why the barotropic ($R_d \rightarrow \infty$) stability predictions remain accurate even at moderate R/R_d : the leading deformation-radius correction is stability-neutral.

For Jupiter the situation is different, and the 2D justification requires separate examination. Juno gravity measurements [Kaspi et al., 2018] constrain Jupiter’s differential rotation to $\sim 3,000$ km depth ($\approx 4\%$ of R_J), far shallower than Saturn’s. Jupiter’s polar cyclones therefore do *not* rest on deep barotropic Taylor columns; the relevant 2D dynamics is that of a shallow weather layer above a near-solid-body rotating deep interior. Two limiting cases bracket the problem:

Deep weather-layer limit ($H_{\text{eff}} \sim 3,000$ km). Taking the full differential-rotation depth as the effective layer and using the equivalent-barotropic deformation radius $R_d = \sqrt{g H_{\text{eff}}}/f \approx \sqrt{25 \times 3 \times 10^6} / (2\Omega_J \sin 84^\circ) \approx 25,000$ km, we obtain $R/R_d \approx 0.3 \ll 1$. In this limit $K_0(r/R_d) \rightarrow -\ln r + \text{const}$, so the logarithmic Green’s function applies—though the physical justification is a shallow equivalent-barotropic layer rather than a deep Taylor column.

Tropospheric limit ($H_{\text{eff}} \sim 30\text{--}100$ km). If the active polar-cyclone layer is shallower, reduced gravity $g' \sim 0.1\text{--}1$ m s $^{-2}$ gives $R_d \sim 2,000\text{--}6,000$ km and $R/R_d \sim 1\text{--}4$. The $N = 8$ ring lies in the finite- R_d regime. Computing the κ_0 -shifted constrained Hessian eigenvalue for the K_0 interaction with $\kappa_0/\kappa = 0.7$ (the Juno central estimate):

N	κ_0/κ	$(R/R_d)_{\text{crit}}$	Tropospheric stable?
6	0.7	17.2	yes ($\gg 4$)
7	0.7	4.2	marginal
8	0.7	0.79	no (< 1)
9	0.7	0.10	no

Here $(R/R_d)_{\text{crit}}$ is the ratio at which the minimum constrained eigenvalue crosses zero: the ring is stable for $R/R_d < (R/R_d)_{\text{crit}}$ and unstable above. For $N = 8$ the critical ratio is $(R/R_d)_{\text{crit}} \approx 0.79$, which lies *below* the tropospheric range $R/R_d \in [1, 4]$. **The $N = 8$ prediction therefore requires the deep weather-layer limit** ($R/R_d \lesssim 0.8$, i.e. $R_d \gtrsim 1.3 R \approx 9,700$ km); it does not survive into the tropospheric regime.

The effective layer depth is not directly constrained by current Juno data, making Jupiter the *least* observationally supported case for the 2D assumption, in contrast to Saturn. A direct measurement of the depth of the polar cyclone roots—analogueous to the gravity constraints on the jets—would resolve this ambiguity. The computation above sets a quantitative target: $R_d > 9,700$ km is necessary for $N = 8$ stability under the K_0 interaction.

The critical ratio is defined implicitly by the equation $\bar{\lambda}(\eta_{\text{crit}}, N) = 0$, where the dimensionless eigenvalue $\bar{\lambda}(\eta, N) = R^2 \cdot \lambda_{\min}(\eta)$ interpolates smoothly from the Havelock value $\bar{\lambda}(0, N) = (N-1) - \lfloor N/2 \rfloor (N - \lfloor N/2 \rfloor)/2$ to zero.

The Lagrange multiplier is determined by the stationarity condition $\partial H/\partial R + \mu_L \partial L/\partial R = 0$. For $H = -\sum_{j < k} K_0(d_{jk}/R_d)$ with $d_{jk} = 2R \sin(\pi p/N)$ and $L = NR^2$, the radial derivative of each pair contributes $-K'_0(d_{jk}/R_d) \cdot 2 \sin(\pi p/N)/R_d = K_1(d_{jk}/R_d) \cdot 2 \sin(\pi p/N)/R_d$ (using $K'_0 = -K_1$). Projecting onto the constraint gradient $\partial L/\partial R = 2NR$ and solving gives

$$-2\mu_L = \frac{\eta}{R} \sum_{p=1}^{N-1} \frac{(1 - \cos \theta_p) K_1(2\eta \sin(\pi p/N))}{2 \sin(\pi p/N)}, \quad \theta_p = 2\pi p/N, \quad (7.1)$$

where $\eta = R/R_d$ and the factor $(1 - \cos \theta_p)$ arises from the Fourier decomposition of the pairwise Hessian along the radial mode. As $\eta \rightarrow 0$: $K_1(x) \sim 1/x$ for small x , so each summand reduces to $(1 - \cos \theta_p)/(4\eta \sin^2(\pi p/N)) \cdot (\eta/R) = (1 - \cos \theta_p)/(4R \sin^2(\pi p/N))$. Summing over p using the Havelock identity $\sum_p (1 - \cos \theta_p)/(4 \sin^2(\pi p/N)) = (N-1)/2$ recovers the flat-plane value $-2\mu_L = (N-1)/(2R^2)$. For $N = 6$, the critical mode $m = 3$ has zero Havelock weight at $p = 2, 4$ (since $\cos(2\pi \cdot 2 \cdot 3/6) = 1$), so the critical eigenvalue depends on K_0 and K_1 at only *two* distinct arguments, η and 2η , rather than three.

No closed-form analog of the Havelock identity is known for K_0 . The orthogonal-ity mechanism that collapses the logarithmic sum to $m(N-m)/2$ relies on the identity $\sum_p e^{2\pi i m p/N} \ln |2 \sin(\pi p/N)| = 0$, which has no counterpart for K_0 : the Bessel values $K_0(2\eta \sin(\pi p/N))$ are transcendental at nonzero algebraic arguments by Siegel's theorem on E -functions, but this does not preclude the full weighted sum from having a closed form. Whether $\sum_p (\cos \theta_{pm}) K_0(2\eta \sin(\pi p/N))$ admits a closed form for any N and m is an open question; no such form has been found, and all stability computations for K_0 interactions are carried out numerically. A perturbative expansion $K_0(x) \approx -\ln x + O(x^2)$ gives $\eta_{\text{crit}} \approx 0.9$ for $N = 6$ —30% below the exact value—because the critical ratio lies in the non-perturbative regime where K_0 departs significantly from $-\ln$. The transition occurs precisely where the exponential decay of K_0 (destroying long-range correlations between non-adjacent vortices) overtakes the logarithmic regime, making the suppression inherently non-perturbative. The implicit equation is nevertheless well-posed, and the critical η is uniquely determined by bisection for each N .

The two phase boundaries. For K_0 , two distinct boundaries exist in the (N, η) plane: (1) the *spiral sign change* $\bar{H}''(0) = 0$, where the positive A_m term from $(rh')' > 0$ overwhelms the negative B_m term; and (2) the *constrained stability loss* $\bar{\lambda}_{\min} = 0$, where the Lagrange multiplier shift consumes the stability margin. The constrained boundary ($\eta \approx 1.29$ for $N = 6$) is always *inside* the spiral boundary ($\eta \approx 3.2$ for $N = 6$) because the Lagrange shift reduces the margin before H'' changes sign. Physically, the polygon requires long-range correlations between non-adjacent vortices. When $\eta \gg 1$ and the K_0 interaction decays exponentially, only nearest-neighbor correlations survive and the polygon loses its energetic advantage.

For comparison, the Cauchy-blob regularization $-\frac{1}{2} \ln(d^2 + 2\varepsilon^2)$ has $H'' < 0$ for *all* ε —the soft-core never flips the spiral sign. The Gaussian-screened interaction $-\ln(d) \cdot e^{-d^2/L^2}$ shows an earlier transition ($\eta \approx 0.9$ for $N = 6$) due to its faster-than-exponential decay.

The constrained stability boundary. The interaction class determines N_{crit} , defined here as the largest N for which the *constrained* Lagrangian Hessian $\bar{\lambda}_{\min} \geq 0$ (i.e. the polygon is a constrained energy minimum). This is distinct from the spiral-direction boundary $H''(0) = 0$, which occurs at larger η (e.g. $\eta \approx 3.2$ vs. 1.29 for $N = 6$ with K_0 ; see the two phase boundaries above):

Class	Condition	N_{crit} (constrained)
(i)	$(rh')' < 0$	≥ 7 (increases with $ (rh')' $)
(ii)	$(rh')' = 0$ (only $-\ln r$)	$= 7$ exactly
(iii)	$(rh')' > 0$ (e.g. K_0)	≤ 7 (decreases with screening)

Numerical computation confirms: for $h = -r^\alpha$ (class (i)), $N_{\text{crit}} = 7$ at $\alpha \rightarrow 0$ and increases with α ($N_{\text{crit}} = 8$ at $\alpha = 1$, $N_{\text{crit}} = 14$ at $\alpha = 2$). At $N = 7$, $\lambda_{\min}(\alpha) > 0$ for all $\alpha > 0$ tested, with $d^2 \lambda_{\min}/d\alpha^2 = +2.5 > 0$ at $\alpha \rightarrow 0$: the marginal stability at $N = 7$

is *unique to the logarithmic interaction*, and any class (i) interaction makes the heptagon strictly stable. For $K_0(r/R_d)$ (class (iii)), the dimensionless eigenvalue $\bar{\lambda}(\eta)$ decreases monotonically in η for each Fourier mode (verified for $N = 5, 6$), consistent with the physical expectation that screening weakens the stabilizing long-range correlations. For $N_{\text{crit}} = 6$ at $R/R_d = 1$ and decreases to 5 at $R/R_d = 3$. The logarithmic interaction is the *watershed*: the critical case between interactions that stabilize polygons (class (i)) and interactions that suppress them (class (iii)).

Proposition 7.1 (Min/max duality). *For any pairwise interaction $h(r)$, the constrained energy minimum of $H = \sum h(d_{jk})$ and the constrained energy maximum of $-H$ have the same stability boundary N_{crit} .*

Proof. The Lagrange multiplier for $-H$ is $\mu_L^{-H} = -\mu_L^H$ (from $-\nabla H = (-\mu_L)\nabla L$). The Lagrangian Hessian $\mathcal{L}_{-H} = -\nabla^2 H + 2\mu_L I = -\mathcal{L}_H$. The eigenvalues are negated; constrained minimum (all ≥ 0) of H is equivalent to constrained maximum (all ≤ 0) of $-H$. $\square$

For gravity ($h = -1/r$, $h' > 0$), the N -gon is a constrained energy maximum. The duality gives $N_{\text{crit}}^{\text{gravity}} = N_{\text{crit}}^{\text{Coulomb}}$. Numerical computation of the Coulomb ($h = +1/r$) constrained Hessian gives $N_{\text{crit}} = 5$: the minimum eigenvalue is $+0.73$ at $N = 5$ and -0.02 at $N = 6$. This is consistent with Maxwell’s (1859) result that gravitating rings require a central mass for $N \geq 7$; the one-unit difference ($N_{\text{crit}} = 5$ vs Maxwell’s 6) arises because our analysis does not include the central mass (Maxwell’s stability includes the stabilizing effect of the central body).

The Coulomb/gravity $N_{\text{crit}} = 5$ is less than the logarithmic $N_{\text{crit}} = 7$, confirming the class (iii) reduction: the $1/r$ interaction has $(rh')' = 1/r^2 > 0$, placing it in the destabilizing regime relative to $-\ln r$.

8 Circulation disorder and stability domain robustness

8.0.1 Resilience to circulation disorder

In both planetary regimes the vortex circulations κ_k are not perfectly equal; convection injects or removes circulation from individual vortices continuously. We model this as $\kappa_k = \kappa(1 + \eta_k)$ with i.i.d. fractional disorders $\eta_k \sim \mathcal{N}(0, \sigma_\eta^2)$.

We work on $\mathbf{H}^2$ rather than the flat plane because the hyperbolic setting simultaneously tests two effects: circulation disorder and curvature stabilization. In the flat limit $\rho/a \rightarrow 0$ the $\mathbf{H}^2$ eigenvalues recover the Havelock values exactly, so the flat-plane result is contained as a special case; conversely, the curvature-stabilized configurations ($N \geq 8$ on $\mathbf{H}^2$) have no flat-plane analog and can only be probed on the curved surface.

On $\mathbf{H}^2$, the disordered Hamiltonian is

$$H_\kappa = - \sum_{j < k} \kappa_j \kappa_k \ln |z_j - z_k| + \frac{1}{2} \sum_j \kappa_j (K_{\text{tot}} - \kappa_j) \ln(a^2 - |z_j|^2), \quad (8.1)$$

where $K_{\text{tot}} = \sum_k \kappa_k$. At $\eta_k \equiv 0$ this reduces exactly to the uniform Hamiltonian H_{hyp} .

Holding the ring positions at the uniform equilibrium (exact to $O(\eta)$; the $O(\eta)$ shift of equilibrium produces only an $O(\eta^2)$ eigenvalue correction), we compute the minimum Fourier-mode eigenvalue $\lambda_{\text{min}}(\eta)$ via finite differences. Monte Carlo over $n = 1000$ realizations estimates $P(\lambda_{\text{min}} < 0)$ as a function of (σ_η, N, ρ) .

Physical range of σ_η . Juno imagery [Adriani et al., 2018, Gavriel and Kaspi, 2021] shows Jupiter’s eight north-polar cyclones have peak wind speeds ranging from ~ 60 to $\sim 100 \text{ m s}^{-1}$ and diameters from $\sim 4,000$ to $\sim 7,000 \text{ km}$. Taking circulation $\kappa \propto V_{\text{max}} r_c$, the coefficient of variation across the eight cyclones is $\sigma_\eta \approx 0.15\text{--}0.25$. Saturn’s hexagonal jet is more uniform (Cassini wind profiles give $\sigma_\eta \lesssim 0.05$ for the azimuthal circulation variation). The physically relevant range is therefore $\sigma_\eta \in [0.05, 0.25]$.

Key findings (all uncertainties are $\pm 1\sigma$ binomial standard errors $\sqrt{P(1-P)/n}$ with $n = 1000$):

- **Stable configurations are robust.** For $N = 6$ at $\rho = 0.5a$, the instability probability remains below 10% ($\pm 0.9\%$) for $\sigma_\eta \leq 0.2$, covering the full physically relevant range.
- **Curvature-stabilized rings ($N \geq 8$) are similarly robust.** At $N = 8$, $\rho = 2a$ (well inside the stable region on $\mathbf{H}^2$), $P < 0.5$ ($\pm 1.6\%$) for $\sigma_\eta \leq 0.1$. At $\sigma_\eta = 0.2$ (Jupiter’s estimated value), P remains below 15%.
- **The $N = 7$ marginal ring is sensitive.** At $N = 7$, $\rho = 0.05a$ (flat-limit regime), $P(\lambda_{\text{min}} < 0)$ increases monotonically with σ_η : $P \approx 0$ ($\pm 0.3\%$) at $\sigma_\eta = 0.1$ and $P > 0$ for $\sigma_\eta \geq 0.5$. The marginal mode ($\lambda_0 = 0$ at $K = 0$) is the first to be destabilized by disorder, as expected. At Jupiter’s estimated $\sigma_\eta \approx 0.2$, a hypothetical $N = 7$ ring would have $P \approx 0$ —the marginal sensitivity manifests only at unphysically large disorder.

Thus the N_{crit} boundary is a *stability domain boundary, not a knife-edge*: configurations comfortably inside the stable region remain stable under the circulation fluctuations observed at Jupiter ($\sigma_\eta \approx 0.2$) and Saturn ($\sigma_\eta \lesssim 0.05$).

Code: `circulation_disorder.instability_probability(N,  $\rho$ ,  $\sigma_\eta$ )` and `disorder_min_eigenvalue` in `src/planetary_polygons/extensions/circulation_disorder.py`; test suite in `tests/test_circulation_disorder.py`.

8.0.2 The regime criterion

The dimensionless number $\beta R^3/\kappa$ is the ratio of two timescales: the Rossby wave period $T_R \sim 1/(\beta R)$ and the vortex turnover time $T_v \sim R^2/\kappa$. When $\beta R^3/\kappa = T_v/T_R \ll 1$, the vortex interactions complete many orbits before the Rossby wave propagates appreciably, so the dynamics is vortex-dominated. When $\beta R^3/\kappa \gg 1$, the Rossby wave dynamics operates faster than the vortex rearrangement, so the wave mechanism dominates:

Regime	$\beta R^3/\kappa$	Bridge
Vortex-dominated	$\ll 1$	Blob convergence (Proposition 6.1)
Wave-dominated	$\gg 1$	Rossby stationarity (equation (6.7))

Jupiter’s north pole. Throughout this regime calculation κ denotes the *Rankine-core* circulation, which integrates only within the solid-body rotation core and sets the characteristic vortex timescale $T_v \sim R^2/\kappa$. The full-cyclone circulation κ_{full} (which includes the outer irrotational skirt) is larger and is the appropriate quantity for computing orbital periods and eigenfrequencies; it is used there instead (Section Paper III, §21.3). The two estimates are numerically distinguished below.

At 84°N , the planetary vorticity gradient is $\beta_J = 2\Omega_J \cos(84^\circ)/R_J \approx 5.1 \times 10^{-13} \text{ m}^{-1}\text{s}^{-1}$ and the cyclone-ring radius is $R \approx 7,470 \text{ km}$. Each Juno-observed circumpolar cyclone

has Rankine core radius $r_c \approx 700$ km and peak wind $V_{\max} \approx 80$ m s⁻¹ [Adriani et al., 2018, Gavriel and Kaspi, 2021], giving $\kappa_{\text{Rank}} = 2\pi r_c V_{\max} \approx 3.5 \times 10^8$ m² s⁻¹ (full-cyclone estimate: $r_c \approx 3,200$ km, $\kappa_{\text{full}} \approx 1.6 \times 10^9$ m² s⁻¹). Hence

$$\frac{\beta R^3}{\kappa} = \frac{5.1 \times 10^{-13} (7.47 \times 10^6)^3}{3.5 \times 10^8} \approx 0.61.$$

This places Jupiter in the vortex-dominated regime, but only moderately: 0.61 is $O(1)$, not $\ll 1$. Rossby-wave effects are not negligible at this value, and a decrease in the effective κ (e.g. using the Rankine-core rather than full-cyclone estimate) would push the ratio above unity. The qualitative conclusion—discrete cyclones, Thomson stability applies, blob bridge proves convergence—is robust to this uncertainty because the vortex-dominated bridge requires only that the cyclones be identifiable as N distinct vortices (which the Juno imagery unambiguously confirms), not that $\beta R^3/\kappa$ be parametrically small. The ratio enters quantitatively only through the Rossby-wave correction to the eigenvalues, which is $O(\beta R^3/\kappa)$ and does not change the sign of $\lambda_{\min}$ for $N = 8$ (where $|\lambda_{\min}| = 1 \gg 0.61$). Using the full-cyclone circulation $\kappa_{\text{full}} \approx 1.6 \times 10^9$ m² s⁻¹ gives $\beta R^3/\kappa \approx 0.13$, firmly in the vortex-dominated regime.

Saturn’s hexagon. At 76°N, $\beta_S = 2\Omega_S \cos(76^\circ)/R_S \approx 1.3 \times 10^{-12}$ m⁻¹ s⁻¹ and $R \approx 14,600$ km. Saturn’s polar jet is well-described by a Gaussian profile $U = U_{\max} \exp(-y^2/2\sigma^2)$; its effective circulation is $\kappa = U_{\max} \sqrt{2\pi} \sigma$. With $U_{\max} \approx 100$ m s⁻¹ and $\sigma \approx 2,100$ km (Cassini VIMS wind profiles), $\kappa \approx 5.4 \times 10^8$ m² s⁻¹, giving

$$\frac{\beta R^3}{\kappa} = \frac{1.3 \times 10^{-12} (1.46 \times 10^7)^3}{5.4 \times 10^8} \approx 7.6.$$

Continuous jet meander, Rossby stationarity selects $n = 6$, periodicity provides quantization.

Both mechanisms require the 2D logarithmic Green’s function as the common mathematical kernel.

9 Predictions for quantum systems: BEC vortex clusters

In a quasi-2D Bose–Einstein condensate in a circular trap, quantized vortices interact logarithmically at scales above the healing length ξ . The circulation is quantized: $\kappa = q \cdot h/m$ with integer winding number q . The κ_0/κ ratio of Proposition 3.1 becomes the winding number ratio q_0/q , yielding sharp integer-valued predictions:

Configuration	κ_{crit}	Prediction
$N \leq 7$ ring	0	Stable without central vortex
$N = 8$ ring	1/2	Unstable alone; stable with $q_0 \geq 1$
$N = 9$ ring	1	Marginally stable with $q_0 = 1$; stable with $q_0 \geq 2$
$N = 10$ ring	7/4	Unstable with $q_0 = 1$; stable with $q_0 \geq 2$

The $N = 8$ prediction is the sharpest: a ring of 8 unit-winding vortices should be unstable alone and stable upon adding a single central vortex ($q_0 = 1 > \kappa_{\text{crit}} = 1/2$).

The $N = 7$ case offers a distinct signature. The quartic restoring force ($\alpha_0 = 45/14$, Section 4) produces an *amplitude-dependent* oscillation frequency for the $m = 3$ mode. In the normal form $\mathcal{H} = (\alpha_0/4)\varepsilon^4$, the nonlinear oscillation frequency is $\omega_{m=3} = \alpha_0 \varepsilon^2 \kappa / (2\pi R^2)$, where ε is the dimensionless perturbation amplitude (radial displacement / ring radius) and $\kappa = h/m_{\text{atom}}$ is the single-quantum circulation. For ^{87}Rb in a circular trap of radius $R = 50 \mu\text{m}$: $\kappa \approx 1.2 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$, $\kappa / (2\pi R^2) \approx 7.6 \text{ s}^{-1}$, so $\omega_{m=3} \approx 24 \varepsilon^2 \text{ s}^{-1}$. At a 5% radial displacement ($\varepsilon = 0.05$): $\omega_{m=3} \approx 0.06 \text{ s}^{-1}$ (period $\sim 100 \text{ s}$), resolvable with standard BEC vortex imaging at $\sim 10 \text{ Hz}$ frame rates. The key signature distinguishing the $N = 7$ quartic confinement from the $N \leq 6$ quadratic regime is that the $m = 3$ oscillation frequency *vanishes* as $\varepsilon \rightarrow 0$ (rather than approaching a finite value), producing an amplitude-dependent period that is directly measurable.

The healing length ξ provides a natural blob width $\varepsilon_{\text{blob}} = \xi/R$. For typical BEC parameters ($\xi \sim 0.3 \mu\text{m}$, $R \sim 50 \mu\text{m}$): $\varepsilon_{\text{blob}} \approx 0.006$, well within the point vortex regime. The blob correction (Proposition 6.1) is negligible: $\lambda(0.006) \approx \lambda_0 + O(10^{-4})$.

Finite-temperature effects. Current BEC vortex experiments operate at finite temperature ($T/T_c \sim 0.3\text{--}0.7$), where a thermal cloud coexists with the condensate and exerts a dissipative drag on vortex motion [Fetter, 2009]. This drag damps the oscillation modes on a timescale $\tau_{\text{drag}} \sim R^2 / (D_v T/T_c)$ where D_v is the vortex diffusion coefficient. For the $N = 7$ quartic mode to be observable, the oscillation period ($\sim 100 \text{ s}$ at $\varepsilon = 0.05$) must be shorter than τ_{drag} ; at $T/T_c = 0.5$ this requires $D_v \lesssim 10^{-9} \text{ m}^2 \text{ s}^{-1}$, which is achievable in large-radius traps ($R \geq 50 \mu\text{m}$) at low temperature. The $N = 8$ stability prediction (unstable alone, stable with $q_0 = 1$) is less sensitive to dissipation because it concerns exponential growth vs. decay, not oscillation frequency; finite temperature adds a damping channel but does not change the sign of the eigenvalue.

The hard-wall disk: an exact theorem. In a hard-wall circular trap of radius R , each vortex at z has an image at $R^2/\bar{z}$. The effective interaction is *exactly* the Green's function on the Poincaré disk of radius R (the conformal equivalence is the identity map).

Proposition 9.1 (Hard-wall BEC = Poincaré disk). *In a hard-wall circular BEC of radius R , the stability eigenvalue of an N -gon vortex ring at radius r is $\lambda_m = C_1(\xi) - m(N-m)/2$ with $\xi = r^2/R^2$ and $C_1(\xi) = (N-1)(1+\xi^2)/(1-\xi)^2$ —the same formula as for $\mathbf{H}^2$ (Paper I, equation 5.7).*

Proof. The image vortex construction gives the Green's function $G(z, w) = -(1/2\pi) \log |z-w| + (1/2\pi) \log |1-z\bar{w}/R^2|$, which is exactly the Green's function on the Poincaré disk of radius R . The Havelock decomposition applies without approximation. $\square$

The palindromic thresholds $\xi^*(N)$ are therefore *exact* predictions for hard-wall BEC experiments:

N	$\xi^* = r^2/R^2$	r/R	Prediction
≤ 7	0	0	Stable at any radius
8	$8-3\sqrt{7} \approx 0.063$	0.25	Stable for $r > 0.25 R$
11	$(1+\sqrt{2})^{-2} \approx 0.17$	0.41	Stable for $r > 0.41 R$
23	$\varphi^{-2} \approx 0.38$	0.62	Stable for $r > 0.62 R$

The **key testable prediction**: an 8-vortex ring at $r/R > 0.25$ in a hard-wall disk BEC should be stable; at $r/R < 0.25$ it should be unstable to the $m = 4$ mode. The threshold $r/R = \sqrt{8-3\sqrt{7}} \approx 0.25$ is an algebraic number in $\mathbb{Q}(\sqrt{7})$ —the same field that governs the $7 \rightarrow 8$ transition in planetary vortex dynamics.

Flat-bottomed BEC traps, in which the point vortex model has been experimentally validated [Gauthier et al., 2019], are the natural setting for testing these predictions.

The harmonic trap: an approximation. For a harmonic trap $V(r) = \frac{1}{2}m\omega^2 r^2$ with Thomas–Fermi radius R_{TF} : the identification $\xi = r^2/R_{\text{TF}}^2$ holds to leading order, but the soft boundary introduces corrections at $O(\xi^2)$ relative to the hard-wall case (the image vortex positions shift by terms proportional to the trap anharmonicity). The prediction table above applies for $\xi \ll 1$ ($r \ll R_{\text{TF}}$), where the hard-wall and harmonic cases coincide. For vortex rings near the condensate edge ($\xi \rightarrow 1$), the harmonic trap softens the effective C_1 , and the point vortex model itself requires corrections from the inhomogeneous density profile.

Healing-length and quantum corrections. The BEC healing length ξ_{heal} sets the blob scale $\varepsilon = \sqrt{2}\xi_{\text{heal}}/d$ where $d = 2r \sin(\pi/N)$ is the intervortex spacing. For typical parameters ($\xi_{\text{heal}} \approx 0.3 \mu\text{m}$, $R \approx 50 \mu\text{m}$): $\varepsilon \approx 0.02$, giving $\delta\lambda \sim \varepsilon^2 \sim 0.04\%$ (Proposition 6.1).

The *quantum* correction (from the Bogoliubov–de Gennes excitation spectrum) is even smaller. The BdG anomalous coupling between vortex cores separated by d is $O(\exp(-d/\xi_{\text{heal}}))$; for $N = 8$ at the threshold ($d \approx 9.6 \mu\text{m}$): $\exp(-d/\xi_{\text{heal}}) \approx 10^{-14}$. The effective mass renormalization shifts the precession frequency but not the energy Hessian (the stability condition $\lambda_m > 0$ is a property of the potential landscape, not the kinetic term). The palindromic threshold $\xi^*(8) = 8 - 3\sqrt{7}$ is therefore an *exact* feature of the quantum spectrum, preserved to exponential accuracy by the topological invariance of the K_0 class (Paper I, Theorem 6.12).

Hyperbolic lattice realisations. Interacting bosons on a $\{p, q\}$ hyperbolic lattice (Bose–Hubbard model, hopping t , on-site U) reduce to the point vortex Hamiltonian in the Thomas–Fermi regime ($U \gg t$): the mean-field condensate $\psi = \sqrt{\rho} e^{i\theta}$ has vortices where $\psi = 0$, and θ satisfies $\Delta_{\text{lattice}}\theta = 2\pi \sum_k \delta(z - z_k)$, giving the lattice Green’s function interaction $H = -\sum_{j < k} G_{\text{lattice}}(z_j, z_k)$. In the continuum limit (lattice spacing $a \ll$ intervortex spacing d): $G_{\text{lattice}} \rightarrow G_{\mathbf{H}^2}$ and the Havelock thresholds apply. The continuum limit requires $\gtrsim 1000$ lattice sites—beyond current circuit-QED (~ 100 sites) but within reach of topoelectric circuits ($\sim 10^4$ sites). On the Bolza tiling ($\{8, 8\}$, genus 2, $|\text{Aut}| = 48$): the prediction is $N_{\text{crit}} = 12$ at the Aut-fixed centre, with $\delta C_1 \approx 0.011$ (Paper I, §7).

Remark 9.1 (Mesoscopic superconducting disks). *Abrikosov vortices in a type-II superconducting disk of radius R obey the same boundary condition as BEC vortices in a hard-wall trap (supercurrent vanishes at the edge $\Leftrightarrow$ Kelvin images at $R^2/\bar{z}$). Proposition 9.1 therefore applies directly: $\lambda_m = C_1(\xi) - m(N-m)/2$ with $\xi = r_0^2/R^2$. The observed experimental sequence in mesoscopic Nb disks (Grigorieva et al. 2006): single polygon for $N \leq 6$, marginal polygon or shell (1+6) at $N = 7$, and shells (1+7, 2+7, ...) for $N \geq 8$, is consistent with $N_{\text{crit}} = 7$ at the disk centre ($\xi = 0$) and central-vortex stabilisation (Section ??) raising N_{crit} to 8 for the outer ring.*

Remark 9.2 (Laughlin quasihole polygons). *The effective Hamiltonian of the Laughlin state at filling $\nu = 1/m$ is $H_{\text{eff}} = -m \sum_{j < k} \log |z_j - z_k| + (4\ell_B^2)^{-1} \sum |z_k|^2$: the logarithmic energy with harmonic confining. Quasiholes (created by factors $\prod (z_j - \eta_a)$) interact via $H_{\text{qh}} = -(1/m) \sum_{a < b} \log |\eta_a - \eta_b| + \text{confining}$, where the $1/m$ comes from the fractional charge $e^* = e/m$. For P quasiholes in a ring at radius r inside a Laughlin droplet of N electrons (radius $R = \ell_B \sqrt{2N}$): the Havelock eigenvalue is $\lambda_p = (1/m)[C_1(\xi) - p(P-p)/2]$ with $\xi = r^2/R^2$. The factor $1/m$ is overall: the stability threshold P_{crit} is independent of the filling fraction m . The palindromic thresholds of Paper I, §5 apply directly: $P \leq 7$ is stable at any r ; $P = 8$ requires $r/R > 0.25$; etc. The Berry phase*

(statistical angle $\theta = \pi/m$, giving anyonic braiding) adds an imaginary term to H_{eff} that affects the dynamics but not the energy Hessian—the stability eigenvalues are real and independent of the statistical angle. The conditions: $N \gtrsim 55$ electrons (for $P = 8$ at $r/R = 0.25$), and $d = 2r \sin(\pi/P) \gg \ell_B$ (well-separated quasiholes).

Remark 9.3 (Non-abelian quasiholes and topological quantum computation). *The Moore–Read state ($\nu = 5/2$) has quasiholes carrying Majorana zero modes with non-abelian braiding. The effective interaction between two quasiholes at separation d is $E(d) = E_{\text{log}}(d) + \delta E_{\text{fusion}}(d)$ where $\delta E_{\text{fusion}} \sim C \exp(-d/\xi_E)$ with $\xi_E \approx 2.3 \ell_B$ (Baraban et al. 2009). The exponential decay gives effective exponent $\alpha = \infty \gg 2$ in the RG classification (Proposition ??): the fusion correction is deeply irrelevant. By the blob bridge (Proposition 6.1): $P_{\text{crit}} = 7$ is unchanged.*

The same argument applies to all gapped quantum Hall states (Read–Rezayi, hierarchy states): the fusion correction is always exponential in d (from the bulk energy gap), hence always RG-irrelevant. $P_{\text{crit}} = 7$ is therefore universal across all gapped quantum Hall states, independent of filling fraction, braiding statistics, and fusion channel.

For topological quantum computation: standard braiding protocols use $P \leq 6$ quasiholes (4 for Moore–Read, 6 for two-qubit Fibonacci), safely below P_{crit} . The threshold is not a practical constraint on current proposals but is a fundamental limit on the complexity of quasihole polygon configurations in any quantum Hall platform.

Remark 9.4 (Independence of spectral and TQFT structures). *On a torus $\Sigma = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$, the topological correction $\delta_m(\Sigma)$ to the Havelock eigenvalue has two natural candidate descriptions:*

1. Geometric spectral theory. *The Laplacian spectrum of Σ (eigenvalues $n_1^2 + n_2^2$ on the lattice) gives a spectral expansion of δ_m . A 20-mode fit achieves $R^2 = 0.79$: the correction is 79% spectral (heat kernel / Selberg type), with the remaining 21% from the full $\log |\theta_1|$ structure.*
2. TQFT fusion labels. *For $N = 23$ at level $k = 5$, the quantum Casimir $f_q(m, N) \in \{0, 1, \varphi\}$ decomposes the modes into three $SU(2)_3$ sectors (null, vacuum, Fibonacci). The Verlinde S -matrix of $SU(2)_3$ has off-diagonal entries ~ 0.6 , which would mix sectors under $\tau \rightarrow -1/\tau$.*

These two structures are independent. The modular S -transform $\tau \rightarrow -1/\tau$ acts as a scalar on all three fusion sectors (ratio $\delta_m(i/2)/\delta_m(2i) \approx 0.97$ for each sector, with spread $< 0.4\%$), not via the Verlinde S -matrix. The fusion labels organise the mode space algebraically, but the spectral corrections do not respect this organisation: within-sector variance equals 95% of the total variance.

The TQFT and the geometric spectral theory share the torus as domain and share the palindromic/modular symmetries, but their spectral data act on different spaces. The fusion identities (Theorem 3 of the q -deformation analysis) describe the algebraic quantisation of the Casimir spectrum; the torus corrections describe the geometric modification of the eigenvalues. The two are complementary, not interchangeable.

10 Gravitational polygon stability in 2+1 dimensions

The 2+1D gravity literature studies the single conical deficit [Deser et al., 1984], two-body scattering, BTZ black holes, polygon tessellations of spatial slices ('t Hooft 1993;

note that “polygon” there refers to the tiles of the tessellation, not to a configuration of point masses), and one-loop partition functions [Giombi et al., 2008]. What appears not to have been asked is: *for how many point masses can one form a stable regular polygon on a spatial slice, and what algebraic number theory governs the answer?* This section derives the answer from the vortex stability framework of Paper I.

In 2+1 dimensional gravity with negative cosmological constant ($\Lambda = -1/\ell^2$), space-time is locally AdS_3 and has no propagating degrees of freedom. A point mass m creates a conical deficit of angle $\alpha = 8\pi Gm$ [Deser et al., 1984]. A constant-time spatial slice is a quotient $S = \Gamma \backslash \mathbf{H}^2$, and the *linearised* gravitational potential of a point mass on S satisfies $\nabla^2 \Phi = 8\pi G m \delta(z - z_0)$ —the same Laplace equation as for the stream function of a point vortex.

Proposition 10.1 (Gravitational polygon stability). *In the linearised regime ($8\pi Gm_i \ll 2\pi$, small deficit angles), N equal point masses of mass m on a spatial slice $S = \Gamma \backslash \mathbf{H}^2$ interact via*

$$E_{\text{grav}} = -(8\pi G)^2 \sum_{i < j} m^2 G_S(z_i, z_j), \quad (10.1)$$

where G_S is the Green’s function of the Laplacian on S (the same G_S appearing in the vortex energy with $\kappa \leftrightarrow 8\pi Gm$). The regular N -gon is a critical point of E_{grav} on the angular-impulse surface, and the Hessian eigenvalues are

$$\lambda_m^{\text{grav}} = C_1(S, \xi) - \frac{m(N-m)}{2}, \quad m = 1, \dots, N-1, \quad (10.2)$$

with the same C_1 and $m(N-m)/2$ as for vortices. The N -gon is a stable equilibrium ($\lambda_m^{\text{grav}} > 0$ for all m) if and only if $N \leq N_{\text{crit}}(S, \xi)$.

Proof. The linearised gravitational potential satisfies the same Poisson equation as the vortex stream function, with $\kappa = 8\pi Gm$. The interaction energy (10.1) is therefore $E_{\text{grav}} = (8\pi Gm/\kappa)^2 \cdot H_{\text{vortex}}$, a positive multiple of the vortex Hamiltonian. At the N -gon, the first and second variations of E_{grav} are proportional to those of H_{vortex} , so the critical points, the Hessian eigenvalues, and the stability criterion ($\lambda_m > 0$ for all m) are identical. The Havelock decomposition (Paper I, §4) gives (10.2).

Note on dynamics. The vortex equations of motion are *first order* (the Hamiltonian is the stream function), while the gravitational equations are *second order* (kinetic + potential energy). The stability criterion—positive-definite Hessian of the potential—is the same for both: a positive eigenvalue $\lambda_m > 0$ gives stable oscillation in the gravitational case and stable circulation in the vortex case. A negative eigenvalue gives exponential growth in both. The criterion $N \leq N_{\text{crit}}$ is therefore identical.

Validity and the developing map. For N equal masses in a ring, each creates a deficit angle $\delta = 8\pi Gm$. The total deficit is $N\delta < 4\pi$ (Gauss–Bonnet for S^2), giving $Gm < 1/(2N)$. We now show that, remarkably, the backreaction of the deficit angles on the flat-plane stability is *exactly zero*.

The **developing map** of the flat cone metric with $\mathbb{Z}_N$ symmetry is

$$f(z) = z \cdot {}_2F_1\left(\beta, \frac{1}{N}; 1 + \frac{1}{N}; z^N\right), \quad \beta = 4Gm, \quad (10.3)$$

with derivative $f'(z) = (1 - z^N)^{-\beta}$. This is a conformal isometry from the cone surface to the Euclidean plane: the metric is $|f'|^2 |dz|^2$, flat away from the cone tips, with cone angle $2\pi(1-\beta)$ at each N -th root of unity.

$\mathbb{Z}_N$ symmetry. $f(\omega z) = \omega f(z)$ for $\omega = e^{2\pi i/N}$: the image of the N -gon of cone points is a regular N -gon in the developed plane, with radius

$$R_f = \frac{\Gamma(1 + 1/N) \Gamma(1 - \beta)}{\Gamma(1 + 1/N - \beta)} \quad (10.4)$$

(Gauss's summation formula for ${}_2F_1$ at $z = 1$). At $\beta = 0$: $R_f = 1$. At the Gauss–Bonnet endpoint $\beta = 1/N$: $R_f = (\pi/N)/\sin(\pi/N)$ (by the Γ -reflection formula). The image radius *increases* with β : the cone metric inflates distances near the tips.

Proposition 10.2 (Conformal invariance of flat-plane stability). *On a flat spatial slice ($\Lambda = 0$) with N equal conical singularities of deficit $\delta = 8\pi Gm$ in a regular ring, the Havelock eigenvalues are*

$$\lambda_m = (N-1) - \frac{1}{2}m(N-m)$$

for all $Gm \in [0, 1/(2N))$. In particular, $N_{\text{crit}} = 7$ is independent of the deficit angle.

Proof. The two-dimensional Green's function is conformally invariant: for any conformal metric $g = e^{2u}|dz|^2$ on the plane, $\Delta_g G = \delta$ gives $G(z, w) = -(2\pi)^{-1} \log|z-w|$ in conformal coordinates, independent of u . Therefore the interaction energy $H = -\sum_{j < k} \log|z_j - z_k|$ is the same in conformal coordinates as on the flat plane.

The $\mathbb{Z}_N$ symmetry maps the N -gon to a regular N -gon of radius R_f in the developed plane. The angular interaction $-\sum_{j < k} \log|e^{i\theta_j} - e^{i\theta_k}|$ is independent of the radius (since $\log R$ drops from the Hessian), and so are the Fourier eigenvalues $m(N-m)/2$ of the tangential Hessian. The constraint curvature contributes $+(N-1)$ independently of R_f , completing the Havelock decomposition. $\square$

The topology change at $\beta = 1/N$. At the Gauss–Bonnet endpoint $Gm = 1/(2N)$, the total deficit exhausts the full curvature budget and the spatial slice *compactifies* to S^2 . On the sphere, the Green's function acquires a curvature correction: $G^{S^2} = -(2\pi)^{-1} \log|z-w| + (4\pi)^{-1} \log[(1+|z|^2)(1+|w|^2)]$ (in stereographic coordinates). The additional term replaces $C_1 = N-1$ with $C_1 = (N-1) \cos \varphi < N-1$, giving $N_{\text{crit}}^{S^2} \leq 6$. The transition from $N_{\text{crit}} = 7$ (flat, all $Gm < 1/(2N)$) to $N_{\text{crit}} \leq 6$ (sphere) is therefore *discontinuous*: it occurs at the topology change, not through a gradual backreaction. The curvature-stability dichotomy of Proposition 10.3 arises from the cosmological constant Λ (which sets $\xi = R^2/\ell^2$), not from the deficit angles.

A remark on “no gravitational force in 2+1D.” Pure 2+1D gravity has no propagating degrees of freedom and no Newtonian $1/r$ force at a distance. But point masses on an AdS_3 spatial slice $\Gamma \backslash \mathbf{H}^2$ do have a well-defined static potential: the Green's function of the Laplacian, which is logarithmic. This potential does not propagate dynamically (there are no gravitational waves), but it determines the equilibrium energy landscape. The stability analysis depends only on the second variation of this static energy, for which the Havelock decomposition applies exactly as in the vortex case. $\square$

Proposition 10.1 transfers the full palindromic hierarchy to gravity:

- $N_{\text{crit}} = 7$ on any spatial slice (universality, Paper I, Theorem 6.8).
- The palindromic thresholds of Paper I, §6 are *gravitational phase transitions*: algebraic-number curvature values at which the stability of the mass polygon changes.
- The five algebraic-integer thresholds ($\xi^*(8) = 8-3\sqrt{7}$, ..., $\xi^*(23) = \varphi^{-2}$) give parameter-free predictions for the stability of mass rings in 2+1D AdS.

10.1 The gravitational prediction

The results of this section yield a concrete, parameter-free prediction for 2+1D gravity.

Proposition 10.3 (Maximum stable polygon in 2+1D gravity). *In linearised 2+1D gravity with cosmological constant Λ ($Gm \ll 1/(4N)$, weak-field regime), the maximum number of equal-mass conical singularities that can form a stable regular polygon on a spatial slice satisfies*

$$\text{sgn}(N_{\text{crit}} - 7) = -\text{sgn}(\Lambda).$$

Specifically: $N_{\text{crit}} = 7$ for $\Lambda = 0$; $N_{\text{crit}} \leq 6$ for $\Lambda > 0$; and $N_{\text{crit}} \geq 8$ for $\Lambda < 0$, with N_{crit} non-decreasing in $|\Lambda|$ (Proposition 5.1). The transitions occur at the palindromic thresholds $\xi^(N)$ of Paper I, §5, with $\xi = R^2/\ell_{\text{AdS}}^2$.*

Proof. The gravitational Havelock decomposition (Proposition 10.1) gives $\lambda_m = C_1(\xi) - m(N-m)/2$. The stability condition $\lambda_m > 0$ for all m , the monotonicity of $C_1(\xi)$, and the curvature-stability dichotomy (Paper I, Proposition 6.9) transfer directly. $\square$

Validity. The weak-field condition and the stability condition are *independent*: Gm controls the deficit angle (and hence the validity of the logarithmic approximation), while $\xi = R^2/\ell^2$ controls the ring position (and hence the stability). At 1% of the maximum mass ($Gm = 0.01/(4N)$): the total deficit angle is $\approx 3.6^\circ$ for all N , well within the logarithmic regime. The constraint structures (angular momentum in gravity, angular impulse in vortex dynamics) are isomorphic to $O(Gm)$: both arise from the $SO(2)$ rotational symmetry of the polygon, and the relativistic corrections to the gravitational angular momentum shift the Havelock eigenvalue by $O(Gm) \sim 1\%$ —negligible against binding eigenvalues of $O(1)$.

10.2 The three-layer decomposition

Proposition 10.4 (Orbifold–Havelock correspondence). *The Havelock eigenvalue on $\mathbf{H}^2$ is exactly the $\mathbb{Z}_N$ orbifold CFT ground-state energy at $c = 12N^2$ with angular impulse constraint:*

$$\lambda_m = -E_0(m; c=12N^2) + \mu_L(\xi), \quad (10.5)$$

where $E_0(m) = -N/2 + m(N-m)/2$ is the twisted-sector ground state and $\mu_L(\xi) = C_1(\xi) - N/2$ is the Lagrange multiplier. The central charge $c = 12N^2$ is determined by matching the mode-dependent coefficient $c/(24N^2) = 1/2$; all curvature dependence enters through μ_L alone. The N^2 scaling follows from the ‘t Hooft structure of the Hamiltonian ($N(N-1)/2 \sim N^2$ pairwise terms); the factor $12 = 24/2$ converts between Virasoro and Havelock normalisations.

Proof. The orbifold ground-state energy $E_0(m) = -c/(24N) + cm(N-m)/(24N^2)$ at $c = 12N^2$ gives $E_0 = -N/2 + m(N-m)/2$. Then $-E_0(m) + \mu_L = [N/2 - m(N-m)/2] + [C_1(\xi) - N/2] = C_1(\xi) - m(N-m)/2 = \lambda_m$. $\square$

Proposition 10.5 (Equivariant Riemann–Roch). *The $\mathbb{Z}_N$ -equivariant spectral index of the vortex Laplacian on the cusped Poincaré disk satisfies*

$$\text{ind}_m = \underbrace{\frac{m(N-m)}{2}}_{c_1\text{-term}} + \underbrace{b(N)}_{\text{Todd term}} + \delta_m, \quad (10.6)$$

where $b(N) = N(N+1)/12 - \log 2 + \log N/(N-1)$ is independent of both m and ρ (verified to 10^{-16} for $N = 3, \dots, 24$). The central charge $c = 12b(N) = N^2 + O(N)$ is the same for every mode m , extracted from the Todd correction alone.

Proof. The Havelock eigenvalue satisfies $\lambda_m + f_m = C_1(\rho) + \delta_m$, and $C_1(\rho) = \log(2 \sinh \rho) + b(N) + O(e^{-\rho})$ (the radial dependence is absorbed by the Green's function). Subtracting $\log(2 \sinh \rho)$: $\text{ind}_m := \lambda_m + f_m - \log(2 \sinh \rho) = b(N) + \delta_m$. Since δ_m is traceless ($\sum \delta_m = 0$ to 10^{-15}) and palindromic ($\delta_m = \delta_{N-m}$), the mode-independent part $b(N)$ is extracted exactly. $\square$

The three-layer decomposition $\lambda_m = C_1 - f_m + \delta_m$ is the *spectral equivariant Riemann–Roch theorem*: f_m is the c_1 -contribution (twist-field dimension), $b(N)$ is the Todd-class correction (vacuum energy $\approx c/12$), and δ_m is the one-loop remainder (the Weyl anomaly). The central charge is not a fitted parameter: it is $c = 12b(N)$, determined by the index theorem.

The Wheeler–DeWitt bridge equation. The single degree of freedom ρ (geodesic radius of the polygon) and the potential $V(\rho) = \lambda_{m^*}(\rho)$ (the critical Havelock eigenvalue) define a minisuperspace Wheeler–DeWitt equation:

$$\left[-\frac{\hbar^2}{2c} \frac{d^2}{d\rho^2} + \log(2 \sinh \rho) + b(N) - \frac{m(N-m)}{2} + \delta_m \right] \Psi = 0, \quad (10.7)$$

with $c = 12N^2$ and $\hbar = 1/\sqrt{c}$. Each term has a definite origin:

Term	Expression	Origin
Quantum kinetic	$-\hbar^2/(2c) \Psi''$	Gravity dynamics (the dynamics of ρ)
Classical potential	$\log(2 \sinh \rho)$	$\mathbf{H}^2$ Green's function
Vacuum energy	$b(N) \sim c/12$	Orbifold CFT vacuum (Todd class)
Matter	$m(N-m)/2$	Twist-field dimension (first Chern class)
One-loop	δ_m	Weyl anomaly (traceless, palindromic)

Three regimes.

1. **Classical** ($c \rightarrow \infty$): the kinetic term vanishes; the constraint reduces to $V(\rho) = 0$, i.e., $\lambda_{m^*}(\rho^*) = 0$ —the palindromic threshold, which is the Einstein equation of the vortex-gravity system.
2. **Semiclassical** (WKB): $\Psi \sim \exp(\pm(i/\hbar) \int \sqrt{2cV} d\rho)$, oscillatory in the exterior ($V > 0$, $\sinh$ -branch) and evanescent in the interior ($V < 0$, $\cosh$ -branch). The continuation $\rho \rightarrow \rho + i\pi/2$ maps between them (§??, Lorentzian paragraph).
3. **Quantum** (near threshold): $\Psi \sim \text{Ai}((2c \coth \rho^*)^{1/3}(\rho - \rho^*))$, smoothing the classical horizon over the universal width

$$\delta\rho = \frac{1}{(2c)^{1/3}} = \frac{1}{(24N^2)^{1/3}} \quad (10.8)$$

(verified to 4 digits for $N = 8, \dots, 20$: $\delta\rho/N^{-2/3} \rightarrow 2^{-1/3}$). At $c = \infty$: $\delta\rho \rightarrow 0$ (sharp classical threshold). At finite c : the horizon has quantum width $O(c^{-1/3})$.

The full development of the gravitational theory—including the polygon–BTZ phase transition, the orbifold CFT at $c = 12N^2$, the derivation of the Einstein equations from polygon stability, and the higher-dimensional extension—is in Paper IV.

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